MULTIPHASE MASTER
Erasmus Mundus

MSC Thesis Report

CFD Study of Flow over Linear and Non-linear Outlet Piano Key Weirs Using InterFoam

Student: Taha Javed

Daily Supervisor: Rabs Ranjit, M.Sc

Academic Supervisor: Prof. Olivier Bonnefoy,

Examiner: Prof. Dr. Nils Rüther,

Host Institute: Technical University of Munich

Host Chair: Chair of Hydraulic Engineering

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Abstract

Piano Key Weirs (PKWs) are the advanced nonlinear hydraulic structures developed to enhance the coefficient of discharge by enhancing the effective crest length in the folded zig zag pattern within the same footprint. Due to slightly complexity of crest shape, flow over PKWs becomes bit complex by the generation of high turbulence zones, vortices formation, high shear stresses generation and jet impact which may eventually cause riverbeds erosion and scouring phenomena. The present thesis is 3D CFD modeling of Linear and Non-Linear Piano key Weirs using Open FOAM. The interface of air-water was modelled using multiphase model called Volume of Fluid (VOF) with the interFoam as solver. The RANS model was incorporated using two equation k-ω SST model. The geometrical and operating conditions were purely based on experimental data and setup. Four cases are analyzed, comprising of two initial flat-bed configurations and two equilibrium scour-bed configurations reconstructed from experimental measurements. The numerical modelling focused on velocity distributions and profiles, turbulence and vortices formation, shear stresses formation and shear stresses profiles comparison among the four cases that how scoured bed can badly affect the downstream channel and near bed hydraulic conditions.

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Finally, I would like to thank everyone who supported and encouraged me throughout the completion of my Master's thesis.

Declaration

I hereby declare that this Master's thesis is my own original work and has been written independently by me. I have used only the sources, references, and materials that are explicitly acknowledged and cited in this thesis.

All ideas, quotations, data, and other material taken from published or unpublished works of other authors have been properly referenced. This thesis has not been submitted, either in whole or in part, for any other degree or qualification.

I further declare that AI-based assistance, specifically myGenAssist, was used solely to improve the language and clarity of the text. All intellectual content, analysis, interpretation, and conclusions presented in this thesis are my own responsibility.

Place, Date: Munich, 23 August 2026
Name: Taha Javed

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List of Abbreviations

Abbreviation	Definition
ADV	Acoustic Doppler Velocimetry
CFD	Computational Fluid Dynamics
GCI	Grid Convergence Index
HPC	High-Performance Computing
LDV	Laser Doppler Velocimetry
LOPKW	Linear Outlet Piano Key Weir
LRZ	Leibniz Supercomputing Centre
NLOPKW	Non-Linear Outlet Piano Key Weir
PIV	Particle Image Velocimetry
PKW	Piano Key Weir
RANS	Reynolds-Averaged Navier–Stokes
SST	Shear Stress Transport
TKE	Turbulent Kinetic Energy
TUM	Technical University of Munich
VOF	Volume of Fluid

Nomenclature

Roman Symbols

Symbol	Description	Unit
B	Streamwise length of the PKW	m
d_{50}	Median sediment particle diameter	m
g	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
H	Upstream hydraulic head over the crest	m
k	Turbulent kinetic energy	$\mathrm{m^2\,s^{-2}}$
L	Total developed crest length	m
P	Weir height	m
p	Pressure	Pa
p_{rgh}	Modified pressure	Pa
Q	Volumetric flow rate (discharge)	$\mathrm{m^3\,s^{-1}}$
U	Velocity magnitude	$\mathrm{m\,s^{-1}}$
W	Overall channel/weir width	m
W_i	Inlet-key width	m
W_o	Outlet-key width	m

Greek Symbols

Symbol	Description	Unit
α	Water volume fraction	–
μ	Dynamic viscosity	Pa s
ν	Kinematic viscosity	$m^2\,s^{-1}$
ρ	Fluid density	$kg\,m^{-3}$
ρ_s	Sediment density	$kg\,m^{-3}$
ρ_w	Water density	$kg\,m^{-3}$
τ_b	Bed shear stress	$N\,m^{-2}$
τ_c	Critical shear stress	$N\,m^{-2}$
θ	Shields parameter	–
ω	Specific dissipation rate	s^{-1}

Chapter 1

Presentation of the Host Structure

1.1 Technical University of Munich

The Technical University of Munich (TUM) is among Germany's foremost research universities and is internationally acknowledged for its excellence in engineering, natural sciences and technology. Founded in 1868, TUM enjoys an excellent reputation for its top-quality education and innovative research as well as close ties to industry and research institutions around the globe.

The university is regularly ranked among the best universities in Europe and offers a multi-disciplinary setting for scientific innovation and practical problem solving. Its research activities span a wide range of engineering disciplines such as civil engineering, environmental engineering, computational mechanics and fluid dynamics.

TUM offers a high-level scientific research environment with state-of-the-art laboratory facilities, advanced computational resources and international collaborations. The university puts a special focus on global challenges in sustainable infrastructure, water resources, renewable energy and environmental protection.

The present master thesis has been carried out at TUM, where advanced computational facilities and expert academic supervision offered an ideal environment for numerical investigations in hydraulic engineering.

1.2 Host Chair and Research Group

This master's thesis was carried out at the Chair of Hydraulic and Water Resources Engineering within the Department of Civil and Environmental Engineering at the Technical University of Munich.

The chair has expertise in various fields such as hydraulic engineering, including flood management, sediment transportation, hydraulic structures, Eco hydraulics using CFD software, so the main objective of this chair is to utilize advanced computational software to enhance the

understanding of complex multiphase flows and optimize hydraulic structures.

The chair integrates experimental investigations with high-fidelity numerical models to investigate the hydraulic behavior in water bodies such as rivers, spillways, dams, canals etc. There are several computational tools which are widely employed such as Open FOAM, Ansys Fluent, Flow 3D to simulate real physics behind multiphase flows, turbulence, shear stresses under real operating conditions.

The state of the art research environment at TUM provides good infrastructure and advanced computational resources and highly skilled and experienced researchers enabling students from different backgrounds to undertake research projects under the supervision of professors.

1.3 Suitability of the Master Thesis at the Host Institution

The Technical University of Munich provides a comfortable environment to conduct research on hydraulic systems by utilizing computational resources and numerical modelling techniques. The chair of Hydraulic and Water Resources Engineering is highly qualified and experienced with experiments and numerical modelling for real world complex problems, making it suitable for the present study

This master's thesis aims to investigate how flow over a Linear and Nonlinear Piano key Weirs behaves in combination with the flat bed and scour bed configuration by incorporating Volume of Fluid (VOF). The present study will perform VOF simulations to investigate complex behavior of air-water interface, and how turbulence affects this flow by mesh generation to help with the simulation and post processing the results against real-world experimental data to make sure they are accurate.

The research work done here matches the Chair's skills in designing water structures, studying how water flows over dams, and using computational tools to model complex systems. Having access to powerful computers such as CoolMUC cluster and experienced academic guidance enabled me to conduct detailed simulations of the CFD modeling of Piano Key Weir.

The knowledge and experience gained through this thesis contributes to modeling water flow in hydraulic structures, which is useful for ongoing research to make spillways work better, predict water flow, and manage water resources in a way that's good for the environment. The knowledge gained from this project supports efforts to improve how we design and operate these structures, which is important for making sure we have enough water and that it's used in a way that's sustainable.

Chapter 2

Introduction

2.1 Background and Motivation

Water resources management is extremely important as they greatly affect our daily life activities. One of the critical and essential hydraulic structures in dams' systems is the spillway which plays a vital role in controlling water flow rate during floods and storing water during rainy seasons to fulfil future needs. Due to global climate changes every year, there is an increase in demand for spillway capacity to pass large water flow rate in extreme flood and rain conditions [5, 4]. Several conventional weirs are unable to sustain large discharge capacity due to the limited crest length with respect to footprint [5]. There are two possible ways to do that either we can modify the spillway structure by increasing its width which is not feasible as it would be economically expensive, or we can increase crest length with the available footprint which is more convenient and economically feasible [5, 4].

In order to maximize the effective crest length, here the concept of non-Linear weirs comes in. Non-linear weirs such as labyrinth weir and piano key weirs (PKWs) comprise of increased crest length within the given width of channel by introducing folding crest patterns [5, 4]. As we know, discharge coefficients greatly depend on the crest length so increasing crest length can accommodate high discharge through the weir without modifying the hydraulic structure [4]. Piano key weirs are more advanced versions of non-linear weirs by incorporating sloped floors and overhangs in upstream or downstream sections of weir depending upon the type of Piano key Weir [4, 6].

Another critical engineering concern is generation of scour at downstream section of Piano Key Weir. When the highly energetic jet strikes the downstream bed, it creates swirling flow called as vortices which initiate the incipient motion of sediment known as scour phenomenon due to shear stresses formation [7, 8]. This scour phenomenon may badly affect stilling basins, riverbeds and hydraulic foundations if not properly addressed [8]. This is the reason that investigating hydraulic behavior and understanding the flow patterns and dynamics over the piano key weir plays an essential role for enhancing spillways and dams structure safety [7, 6].

In the past, the hydraulic behavior and performance of Piano key weirs have been investigated by conducting experiments in a laboratory [6]. Performing experiments can provide us accurate and trustworthy results for the quantities such as upstream water head, discharge coefficient [6]. However, as concerned to other critical quantities associated with the turbulence quantities such as turbulent kinetic energy, velocity and pressure fields, turbulence energy dissipation, vortices formation is practically not possible to measure in real experiments. Obtaining such information generally requires sophisticate and expensive measurement techniques such as Particle Image Velocimetry (PIV), Laser Doppler Velocimetry (LDV) or Acoustic Doppler Velocimetry (ADV), which are not available in many hydraulic laboratories.

Numerical Modelling techniques, especially CFD, have become a robust and efficient tool for addressing complex flow behaviors [6]. After validation of numerical models against reliable empirical data, numerical models can provide spatial and temporal information. Numerical model delivers in depth knowledge of velocity components, pressure fields, turbulent kinetic energy, Reynolds stresses, shear stresses, air-water interface evolution, recirculation zones and jet behavior, which are impractical to extract through experiments [6]. Consequently, Numerical modelling offers cost-effective and robust solutions where physical model testing is expensive or not feasible under some circumstances. Recent studies have therefore increasingly combined physical experiments with validated numerical simulations to obtain a more comprehensive understanding of Piano Key Weir hydraulics [6].

The current study is based on combined experimental-numerical approach. A 3D numerical model (CFD) is developed and validated against reliable empirical laboratory measurements and after the successful validation, model was utilized to investigate in depth downstream hydraulic such as evolution of swirling flow patterns, turbulence energy dissipation, turbulent momentum transport, and shear stresses for both linear and nonlinear geometry of piano key weirs. In short, numerical models with the combination of experimental data can efficiently support us in addressing the causes of scour initiation in riverbeds, dams, hydraulic structures up to significant level.

2.2 Hydraulic Structures and Spillways

Hydraulic Structures are essential for control, regulating, storing and measuring the flow of water in both natural and artificial water bodies. These structures protect us from floods and facilitate us in hydropower generation, water supply for both domestic and commercial uses and environment management. Design of the hydraulic structures plays a critical role in safety and water management under extreme circumstances while maintaining structural stability.

Spillways are the critical hydraulic structures for preserving the safety of storage systems such as dams and huge reservoirs. These Spillways can release excessive water from the upstream channel towards downstream channel in the events of abundant rainfalls or during flooding conditions when the water level exceeds the threshold capacity of dams and storage systems

so that's why studying the hydraulic behavior of dams and spillways is directly associated with safety of hydraulic storage systems.

A typical spillway comprises of various components including approach domain, crest of spillway, downstream channel and energy dissipation systems such as stilling basins etc. A schematic representation of the flow over a spillway and the associated downstream energy-dissipation process is shown in Figure 2.1 (Roushangar et al., 2019). Water exceeding the threshold capacity of dams is allowed to flow over the spillway crest and then water propagates towards the downstream channel. During this process, potential energy of water is converted into kinetic energy resulting in highly turbulent flow and vortices formation and then ultimately the shear stresses formation which is the starting point of scouring process for the riverbeds. To avoid this, energy dissipation systems are installed to reduce the flow aggressiveness before reaching the natural water channel and hence reduces the scouring of downstream and riverbeds.

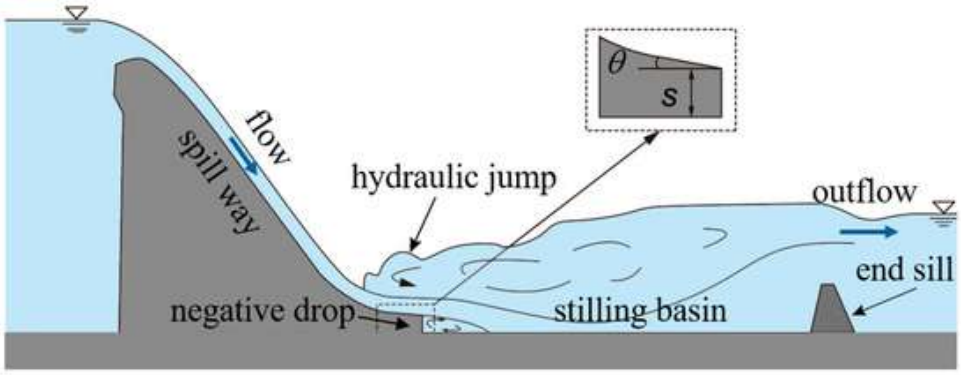


Figure 2.1: Schematic representation of flow over a spillway and the associated downstream hydraulic jump and energy-dissipation process. Adapted from [1]

Several spillways have been developed, and each structure is designed for specific requirements, specific head conditions and so on. Traditional spillways include side-channel spillways, shaft spillways, ogee spillways, chute spillways siphon spillways, and free overflow weirs. Each of these spillways are designed for specific criteria which includes flood magnitude, reservoir capacity, discharge capacity, site location, foundation circumstances and so on. Increasing discharge capacity accounts for increasing spillway width or elevating the reservoir level and both are economically not feasible.

Free overflow is the widely used type of weir in spillway structures to control and regulate water flow rate. Free overflow weirs discharge capacity heavily depends on the crest length and total water head above the crest. Conventional linear weirs provide relatively limited discharge capacity because the crest extends only across the channel width. As a result, increasing the discharge capacity generally requires increasing either the crest length by widening the spillway or increasing the upstream water level. These limitations have motivated the development of non-linear spillway geometries capable of providing substantially greater discharge capacity within the same channel width.

To overcome the limitations of conventional linear weirs, folded spillway geometries such as

labyrinth weirs and Piano Key Weirs (PKWs) have been developed. By folding the crest into a three-dimensional configuration, these structures significantly increase the effective crest length while occupying nearly the same channel width. Since the discharge over a free overflow weir is directly proportional to the effective crest length, non-linear weirs can convey substantially larger flow rates than conventional linear weirs without requiring major modifications to existing dam structures. Consequently, Piano Key Weirs have become an attractive solution for increasing spillway capacity during the rehabilitation and upgrading of existing dams where construction space is limited.

Because of their superior hydraulic efficiency and compact geometry, Piano Key Weirs have attracted considerable attention in recent years for both experimental and numerical investigations. The following section therefore introduces the development, geometry, hydraulic behaviour, and operating principles of Piano Key Weirs in greater detail.

2.3 Linear and Non-linear Weirs

Weirs are overflow hydraulic structures widely used to regulate water levels, measure discharge, and safely pass excess flow in rivers, canals, reservoirs, and dam spillways. The discharge capacity of a conventional weir mainly depends on the crest length and the hydraulic head acting above the crest. Consequently, increasing the crest length is one of the effective methods for increasing discharge without raising the upstream water level. In conventional linear weirs, the overflow crest extends predominantly in a straight line across the channel width. Consequently, the available crest length is largely constrained by the width of the channel or spillway, which limits the discharge capacity so Piano Key Weirs can increase discharge capacity and efficiency by enhancing crest length instead of increasing the structure width or upstream head [5].

To overcome the geometric limitations of linear weirs, non-linear weirs were introduced. Rather than increasing the spillway width, these structures fold the crest in plan view, creating a substantially greater developed crest length within the same available channel width [9]. Since the discharge over a free overflow weir is strongly influenced by the developed crest length, this configuration can significantly increase the discharge capacity without requiring a corresponding increase in spillway width. Labyrinth weirs and Piano Key Weirs (PKWs) are important examples of non-linear weirs [9]. As a result, non-linear weirs have become attractive alternatives for both new hydraulic structures and the rehabilitation of existing spillways where widening the spillway is impractical [9].

Typical configurations of linear and non-linear weirs are illustrated in Figure 2.2.

Numerous geometric parameters influence the hydraulic performance of non-linear weirs, including the sidewall angle, crest shape, crest thickness, key width, number of cycles, and the ratio between the hydraulic head and weir height. Recent studies have also shown that hydraulic losses and energy dissipation vary considerably with weir geometry. Therefore, optimizing the geometric configuration is essential for achieving high discharge efficiency while maintaining

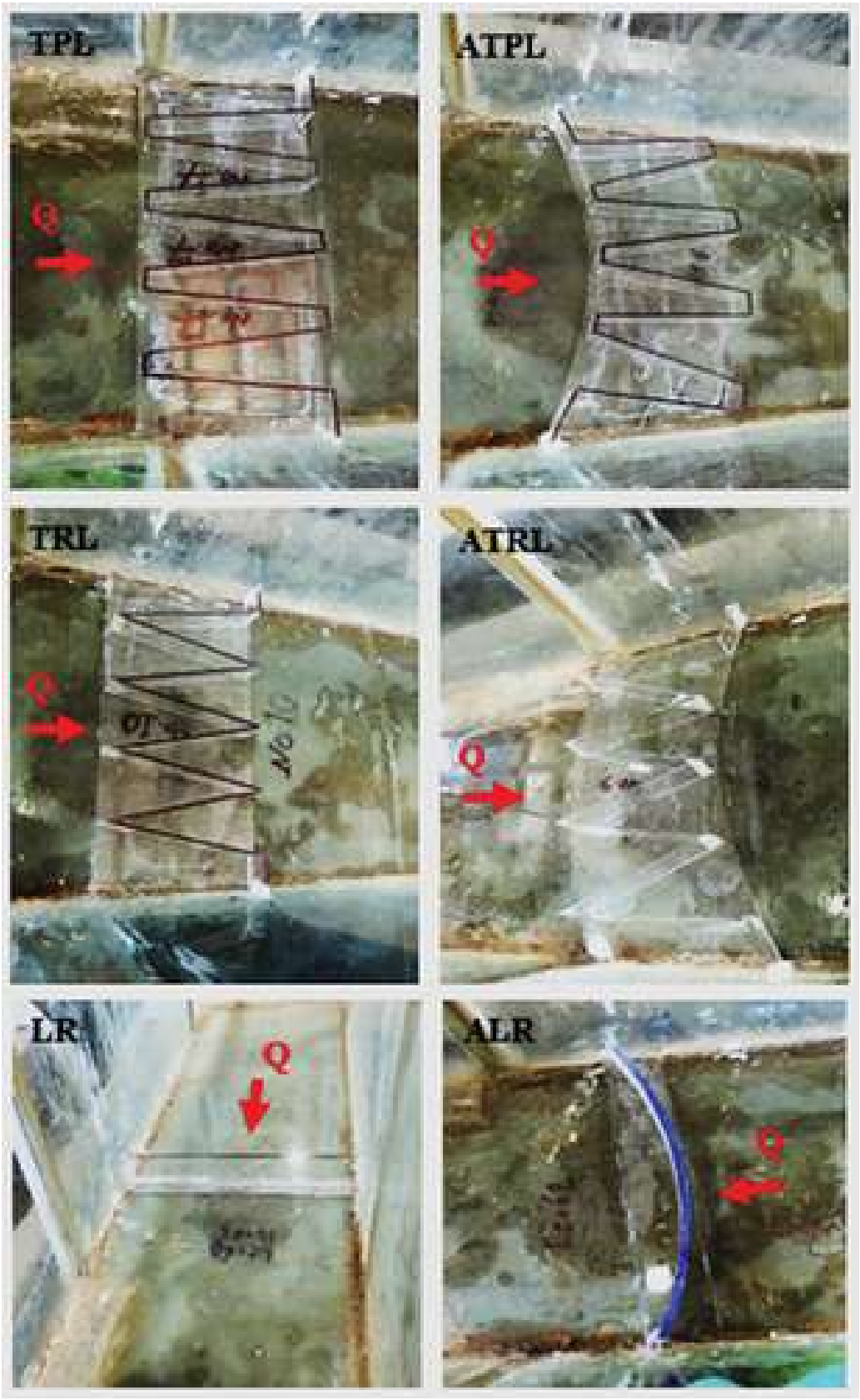


Figure 2.2: Physical models of linear and non-linear weir configurations. Adapted from [2].

stable flow conditions [2].

2.4 Piano Key Weirs

The Piano Key Weirs is an advanced version of nonlinear overflow weir designed to address the problem of geometric and hydraulic limitations of traditional labyrinth weirs [9]. The concept of PKW was first given by Lemprière and Ouamane in 2003 as a compact hydraulic structure capable of allowing higher discharge to pass through within same footprint [9]. Unlike traditional weirs, Piano key Weirs comprises of alternating inlet and outlet keys with upstream and downstream overhangs which increases the effective crest length and thus increasing discharge capacity [9]. The geometry of PKW resembles the keys of piano when PKW viewed from top that's why it is named as Piano key Weirs. The principal geometrical parameters which effect the efficiency and hydraulic performance of PKW are upstream head (H), the height of weir (P), Total crest length (L_d), the widths of the inlet and outlet keys (W_i and W_o), Length of Keys, the slope of the sidewall, the dimensions of the overhang and the head-height ratio (H/P) [9]. These parameters are shown in Figure 2.3.

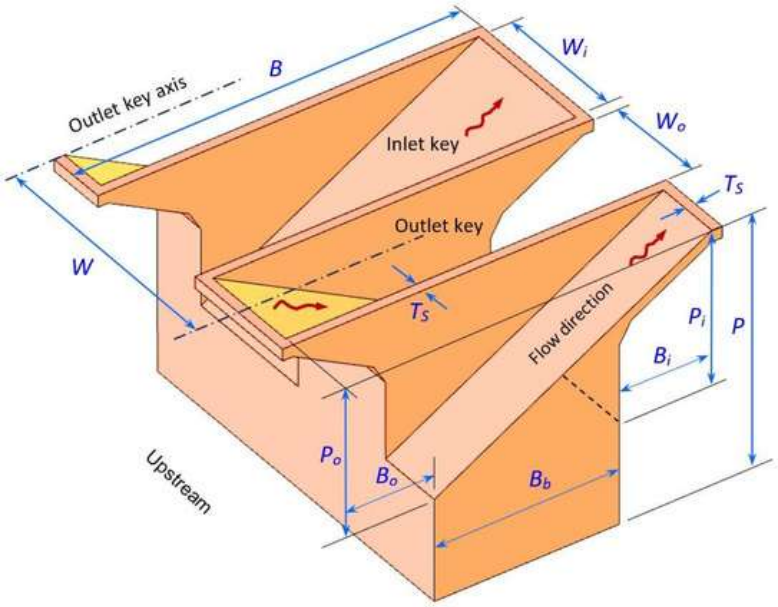


Figure 2.3: Geometric parameters of a Type-A Piano Key Weir. Adapted from [3].

The hydraulic efficiency of PKW is significantly described by 3D flow patterns. As upstream water approaches the PKW structure, it is directed towards the inlet keys before it flows over the crest surface of outlet keys resulting in highly turbulent and aerated flow [10]. At low upstream head, the PKW produces higher discharge capacities as compared to traditional linear weirs and in contrast at higher heads, interaction between napes increases and local submergence conditions reduces the hydraulic efficiency of PKW [5, 10]. PKW are classified based on geometric orientation such as Type A, Type B, Type C, and Type D. Type A contains overhangs

in both upstream and downstream, Type B contains only upstream overhangs, Type C comprised of downstream overhangs [**?**]. This Classification is best illustrated in Figure 2.4. The selection choice of specific type PKW depends upon the structural constraints, hydraulic requirements, desired discharge value, and available construction space [5].

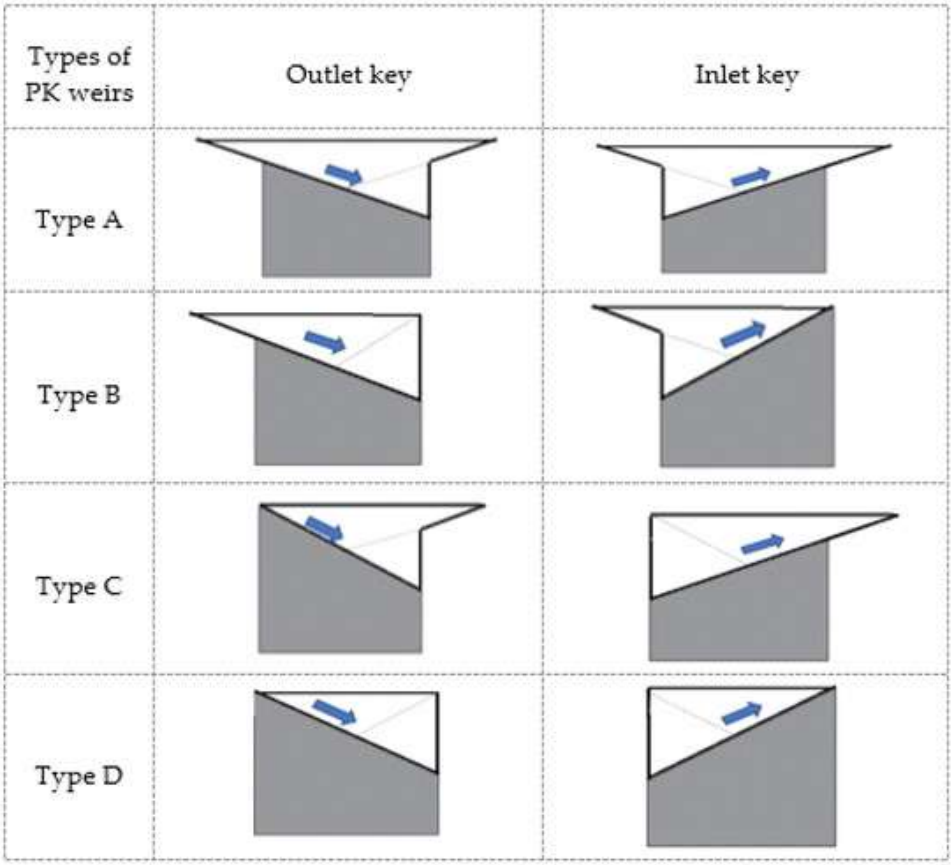


Figure 2.4: Classification of Piano Key Weirs according to the arrangement of upstream and downstream overhangs: Types A, B, C, and D. Adapted from [4].

During the last two decades, PKWs have become convenient solution for newly hydraulic projects and the modification of existing hydraulic structures especially where space constraints prevail [5]. The ability of PKWs to transmit higher discharge within same footprint attracted the attention of several researchers and encouraged them to investigate the hydraulic mechanism of PKWs by conducting experiments and developing numerical models to predict the geometric parameters such as velocity fields, turbulence mixing, vortices and eddies generation and other flow characteristics which are difficult or impossible to investigate experimentally [6, 11]. Numerical approaches such as Reynolds-Averaged Navier–Stokes (RANS) equations with the combination of Volume of Fluid (VOF) methods have become efficient tool for investigating and optimizing the hydraulic performance of Piano Key Weirs [12].

Chapter 3

Literature Review

3.1 Overview

Since the introduction of Piano Key Weirs (PKWs), considerable research has been conducted to improve the efficiency and hydraulic performance of PKW and to better understand the complex flow behavior around the PKW. Majority of the Previous numerical and experimental studies strongly focused on discharge capacity, energy dissipation over PKW and geometric parameters affecting discharge capacity of PKW using several computational tools for CFD to predict flow dynamics and important turbulence parameters.

This chapter highlights the experimental and numerical studies relevant to the present investigation. Special attention and focus were given to experimental and numerical studies concerned about Linear Outlet Piano Key Weirs (LOPKWs) and Non-linear Outlet Piano Key Weirs (NLOPKWs), downstream behavior, bed shear stresses, sediment transport and scour related flow mechanisms.

3.2 Experimental Studies on Piano Key Weirs

Several experimental investigations were carried out by different researchers at different times which have played a vital role in demonstrating the discharged characteristics and flow behavior around PKW. Li et al. [6] performed a physical model test on PKW of Type-A under different upstream head conditions and they found that how the complex three-dimensional flow developing within inlet and outlet keys greatly effects the overall hydraulic performance of PKW

Ghanbari and Heidarnejad [13] experimentally made comparison between rectangular and triangular PKW and combined experimental approach with the numerical modelling using FLOW-3D software. They found experimentally that coefficient of discharge for the triangular PKW configuration was found to be 25% more than that of rectangular PKW configuration. Furthermore, they found that modifying the PKW from a rectangular shape configuration to triangular notch configuration significantly enhance the coefficient of discharge with the magni-

tude of improvement depending on the weir geometry and the ratio of cycle width to weir height (w/P).

More recently, Mishra et al. [14] experimentally demonstrated and investigated the hydraulic performance of Type A PKW configuration by utilizing both experimental findings and numerical modelling. The original model PKW-L employed straight linear profile for both inlet and outlet keys whereas other three models employed curvilinear outlet key profile like ogee-shaped profile at different head to weir height ratio H_d/P. What they found interesting is that curvilinear ogee-shaped outlet keys improved the overall hydraulic performance when the upstream head was equivalent or greater than the design head of PKW. The ogee-shaped outlet keys enable flow to leave more smoothly resulting in less flow separation and faster evacuation and reduced interference of nappe and reduced submergence conditions. The maximum increase in discharge coefficient was approximately 9%, observed for the PKW-$CL_{0.25}$ model [14].

The above studies show that experimental findings play a vital role in investigating and demonstrating the PKW discharge efficiency at varying heads and assessing the influence of geometric parameters before implementing the final practical design. [6, 13, 14].

3.3 Influence of Geometric Parameters on Hydraulic Performance

The overall hydraulic efficiency and performance of a Piano key weir is greatly affected by several geometric parameters including ratio of inlet and outlet key widths, total crest length, overhang length, height of weir and inlet and outlet keys slopes [4, 6].

Bhukya et al. [4] studied and reviewed previous experimental studies and discharge formulation and concluded that the coefficient of discharge depends on the crest-length ratio L/W, upstream-downstream length-weir height ratio B/P. The review further highlighted other important geometric parameters and dimensionless ratios such as inlet and outlet width ratio W_i/W_o, weir height P, and the effect of dimensions of upstream and downstream overhangs on the coefficient of discharge of PKWs [4].

Li et al. [6] combined experimental findings with numerical model to investigate the hydraulic characteristics and coefficient discharge for Type-A PKW under different upstream head conditions and reported that discharge capacity generally decrease with an increase of upstream head H/P due to enhanced flow separation at inlet keys, therefore reducing the effective flow area and thus reducing discharge capacity and efficiency. It was also observed and reported that at higher heads, napes interference increases submergence at outlet keys at downstream region resulting in reduced discharge efficiency.

The shape of the keys can also significantly alter hydraulic performance. Ghanbari and Heidarnejad [13] showed that triangular modifications produced larger discharge coefficients than the corresponding rectangular configuration. Similarly, Mishra et al. [14] found that

curvilinear outlet-key profiles improved discharge efficiency as compared with conventional linear profiles, particularly above the design head.

Overall, the literature indicates that PKW performance is governed by the combined effect of several geometric ratios rather than by a single parameter, making geometric optimization an important aspect of PKW design [4, 14].

3.4 Numerical and CFD Studies

Due to Complex 3d flow natures of Flow around a PKW, it is quite challenging to analyze some important hydraulic quantities experimentally. Numerical models reveal important and extensive information on recirculation zones, velocity fields, streamlines, and vorticity. [6, 15].

Hu et al. [15] incorporated volume of Fluid (VOF) multiphase model to numerically investigate the hydraulic behavior of flow over PKW under different upstream head conditions. They found the reduction in discharge efficiency with the increase in upstream head conditions due to increasing submergence conditions at outlet keys, interference between adjacent napes, changing direction of flow over side crests [15].

Li et al. [6] combined physical model of type A PKW with 3D numerical model to investigate the hydraulic behavior of Type-A PKW. The numerical model revealed extensive and detailed information for Turbulence kinetic energy, streamlines, and velocity fields which enable researchers to analyze complex flow behavior, which is challenging to investigate experimentally [6].

Ghanbari and Heidarnejad [13] also did numerical modelling of flow over triangular and rectangular PKW and combined experimental findings with numerical modelling and they found the outcomes of numerical model were close enough with experimental results. Mishra et al. [14] also combined laboratory findings with RANS-based numerical modelling outcomes for both linear and ogee-shaped curvilinear outlet-key profiles and thus, numerical model provided deep insights on flow vortices, velocity fields, nappes interferences around both linear and ogee-shaped curvilinear PKW.

The accuracy and trustworthiness of CFD results are strongly dependent on the mesh resolution. Bremer and Oertel [16] performed mesh independence study using the Grid Convergence Index (GCI) approach and the outcome revealed that coarse mesh was unable to capture true physics whereas refined mesh was able to capture true physics by giving accurate results for local velocity magnitudes so study therefore highlighted the importance of mesh convergence study to obtain precise and accurate results for PKW flow [16].

The above studies demonstrate that numerical modelling is particularly effective when combined with experimental validation, as laboratory measurements provide confidence in the numerical solution while CFD offers detailed access to the three-dimensional flow field [6, 16].

3.5 Downstream Flow Behaviour and Energy Dissipation

Several previous studies also reported and investigated the hydraulic behavior and energy characteristics associated with piano key weirs and rectangular labyrinth weirs. The outcome revealed that the sloping and stepped outlet key exhibit smoother discharge for near bed streamlines, while the stepped outlet increased the turbulence and aeration in the downstream region. The study also showed that increasing upstream head H/P drastically reduced the energy dissipation and smaller inlet-outlet key width ratio W_i/W_o enhanced the energy dissipation downstream Mirkhorli et al. [17]

The same study reported that at high upstream heads H/P, the energy dissipation decreases nonlinearly because of submergence condition which reduces the nappe interference effect at downstream region resulting in reduced energy dissipation [17].

Belay and Oertel [18] experimentally investigated downstream two-phase flow for an A-type PKW equipped with flat-top, half-round and quarter-round crest shapes. Their measurements showed that crest shape had only a limited effect on the depth-averaged void fraction and particle count rate, whereas the flat-top crest produced stronger flow separation and a larger jet drop height [18].

Belay and Oertel [18] also observed that the downstream air-water interface profile followed a damped undular pattern and that the highest wave crest occurred downstream of the jet-impact region. These results demonstrate that the region downstream of a PKW contains complex three-dimensional and two-phase flow structures that are relevant to downstream hydraulic loading and channel-bed protection [18, 17]. “‘

3.6 Sediment Transport, Bed Shear Stress and Scour

The interaction between Piano Key Weirs and sediment transport has received increasing attention because the highly three-dimensional flow passing through and over the structure can generate hydraulic forces capable of mobilising bed material [7]. Bed shear stress represents the tangential force exerted by the flowing water on the bed surface and is widely regarded as one of the principal hydraulic parameters controlling sediment entrainment, transport, erosion and local scour [8]. Sediment motion begins when the applied bed shear stress exceeds the critical resistance of the particles, with the threshold depending on sediment size, shape, packing and local hydraulic conditions [7].

The initiation of sediment transport is commonly described using the dimensionless Shields parameter,

$$\theta = \frac{\tau_b}{(\rho_s - \rho_w)\, g\, d_{50}}, \tag{3.1}$$

where θ is the dimensionless Shields parameter, τ_b is the bed shear stress (N/m^2), ρ_s is the

sediment density (kg/m^3), ρ_w is the water density (kg/m^3), g is the gravitational acceleration (m/s^2), and d_{50} is the median sediment diameter (m). When the applied bed shear stress exceeds the corresponding critical condition, sediment particles may begin to move by rolling or sliding and may eventually enter the saltation regime, depending on the magnitude of the hydraulic forcing [7].

An early PKW-specific investigation of near-bed turbulence and shear stress was presented by Tiwari and Sharma (2015). Using three-dimensional Acoustic Doppler Velocimeter measurements upstream of a Piano Key Weir, they compared three methods for estimating bed shear stress: the Reynolds-stress method, a turbulent kinetic energy method and a modified turbulent kinetic energy method [19]. Their results showed that the estimated shear stress depended strongly on the selected formulation because the hydraulic structure generated complex three-dimensional turbulence. They found that the turbulent kinetic energy approach provided the most consistent results and appeared to be the most promising method, while the modified turbulent kinetic energy approach produced results similar to the turbulent kinetic energy method as the influence of the hydraulic structure decreased [19]. This study illustrates one of the principal experimental challenges associated with PKWs: highly turbulent and spatially varying flow makes local bed-shear-stress measurements considerably more difficult than measurements in approximately uniform open-channel flow [19].

Kumar et al. (2021) investigated sediment movement over three Type-A PKW configurations using laboratory experiments, high-speed particle tracking, Acoustic Doppler Velocimetry and CFD [7]. Their study considered quartz coarse sand and gravel particles with sizes ranging from approximately 1.7 to 6.3 mm and related particle kinematics to the local hydraulic conditions around the inlet keys [7]. The experiments showed that sediment particles generally decelerated while approaching the inlet key but subsequently accelerated over the key as a result of flow contraction and increasing shear stress [7]. Rolling and saltation regimes were observed, while CFD calculations indicated a rapid increase in shear stress toward the downstream end of the inlet key [7]. For the investigated configurations, approximately 17%–43% additional upstream bed shear stress was required to transport sediment over the PKW, demonstrating that the structure can substantially modify the hydraulic threshold governing particle movement [7].

The same investigation also illustrates why numerical modelling becomes valuable for shear-stress analysis. Kumar et al. measured upstream bed shear stress from ADV velocity profiles using the von Kármán-Prandtl logarithmic velocity relationship [7]. However, direct measurements over the inlet-key bed were difficult because of strong turbulence, unsteadiness, vortices and flow complexity. Consequently, the local shear stress over the inlet key was obtained using CFD and subsequently correlated with the observed sediment motion [7]. This limitation of experimental measurements provides an important motivation for using CFD to determine the spatial distribution of shear stress over complex PKW geometries.

Scour downstream of a hydraulic structure is primarily generated by the interaction between the high-velocity overflow jet and the erodible bed. Jet impact can generate bed-striking flow,

roller vortices and recirculation zones that produce highly non-uniform hydraulic loading and initiate sediment removal [8]. Under clear-water conditions, the initial stage of scour is generally characterised by rapid bed erosion near the jet-impact region. As the scour hole deepens and extends downstream, the geometry of the bed modifies the flow field, progressively reducing the energy of the jet reaching the bed and decreasing the erosion rate [8]. This interaction indicates that scour is not controlled only by the initial hydraulic conditions; the developing bed morphology itself alters the subsequent velocity, turbulence and shear-stress distributions [8].

Recent physical-model investigations have further demonstrated the importance of bed shear stress as an indicator of scour development. Kiraga et al. (2026) examined the relationship between several hydraulic parameters and maximum scour depth and found that bed shear stress exhibited the strongest relationship, with a Pearson correlation coefficient of approximately $r = 0.93$ [8]. In comparison, the velocity measured 1 cm above the bed showed only a weak correlation of approximately $r = 0.26$, while mean velocity and Froude number showed moderate correlations [8]. The authors attributed the stronger performance of bed shear stress to the fact that it directly represents the hydraulic force acting on sediment particles, whereas velocity measurements can be strongly influenced by local turbulence and instantaneous vortical structures [8].

3.7 Research Gap

Previous numerical studies and experiments of Piano Key Weirs highlight the discharge efficiency as the main objective of studies and how discharge efficiency can be increased by changing geometric parameters of PKW.

Few studies were focused on investigating the downstream hydraulic behavior of scour development and how bed shear stresses are generated at downstream bed and the what is the intensity of local bed shear stresses at different location at downstream bed and how the scouring development initiates and what are driving forces causing the scouring process.

The hydraulic process at downstream such as scour developments, bed shear stress formation, velocity field distribution, vortices generation, turbulent kinetic energy distribution are difficult to investigate in experiments. A numerical model simulation is needed to account for these hydraulic quantities.

Furthermore, only a limited number of studies have made comparisons for the hydraulic behavior for Linear Outlet Piano Key Weirs and Non-linear Outlet Piano Key Weirs for flat bed and scoured bed configuration at downstream region.

Therefore, there remains a need for detailed CFD simulations capable of providing high-resolution information regarding velocity distributions, turbulence, pressure fields and bed shear stresses for both linear and non-linear outlet Piano Key Weirs under different downstream bed configurations.

3.8 Research Objectives

The main objective of this research is to numerically analyze and investigate the hydraulic behaviour of flow over over Linear Outlet Piano Key Weirs (LOPKWs) and Non-linear Outlet Piano Key Weirs (NLOPKWs) using Multiphase model called Volume of Fluid Model (VOF) solver `interFoam` available in OpenFOAM.

The specific objectives of this research are:

- To develop accurate 3D CFD model of Linear Outlet Piano Key Weirs and Non-linear Outlet Piano Key Weirs using InterFoam.
- To investigate four numerical model cases consisting of two Piano Key Weir geometries and two downstream bed configurations.
- To investigate velocity distributions, Shear Stresses, turbulent kinetic energy and downstream jet behaviour.
- To calculate and analyze bed shear stress distribution at inlet keys, outlet key and side walls along the the downstream channel and identify prominent regions where the calculated shear stress exceeds the critical shear stress for sediment motion.
- To compare the downstream hydraulic flow behavior of Linear Outlet Piano Key Weirs and Non-linear Outlet Piano Key Weirs under identical hydraulic operating conditions.
- To compare the hydraulic flow behavior for the initial flat-bed orientation and the equilibrium scour-bed orientation.

Chapter 4

Methodology

4.1 Research Methodology and Workflow

The approach and methodology adopted in this study integrates 3D numerical model with experimental data to investigate the hydraulic behavior of flow over nonlinear weir such as Piano Key weirs (PKW). Numerical models were completely based on experimental data and condition, for instance experimental geometry, upstream and downstream water, flat bed and equilibrium scour bed profiles.

The numerical simulation was performed using Open FOAM. The computational domain dimensions were adopted according to literature reviews proportion and computation domain was generated using background blockMesh and then applying snappyHexMesh. Initially the background blockMesh was incorporated first into overall domain and after that snappyHexMesh was incorporated into regions around 3D PKW and provided additional refinement regions to sensitive areas where resolving velocity gradients, vortices, turbulence is essential for us, including nappe region, initial downstream region, scour region.

Four numerical configurations were analyzed. First two cases represent the flat rigid-bed configuration, while the other two cases represent the scoured bed configuration which is entirely based on experimental data. The numerical model was simulated using interFoam solver (Multiphase Solver), and turbulence was resolved using two equation RANS k–ω SST model.

After mesh generation, boundaries and initial condition were adopted based on combination of literature review studies and experimental conditions. The simulation was executed until overall hydraulic flow depicts stable behavior. Post processing of result was focused on velocity distribution, coefficient of discharge, bed shear stress, turbulent kinetic energy, Streamlines, vorticity especially in the downstream region.

4.2 Experimental Configuration and Data

The numerical model was developed using the geometry and hydraulic conditions obtained from the corresponding laboratory investigation of the Piano Key Weir. The experimental channel had a width of 0.82 m, while the PKW had a streamwise length of approximately 0.424 m and a crest height of 0.17 m from the raised bed. The experimental data included discharge, upstream and downstream water levels, Total crest length and downstream equilibrium scour profiles measured after sediment-bed deformation.

The numerical model was produced in accordance with the experimental data obtained from the corresponding laboratory where the experiment was conducted. The experimental channel was 0.82 m wide. An upstream length of 1.50 m was adopted which corresponds to approximately $9P$ to allow the flow to fully develop before it reaches PKW. Downstream length of 3.396 m was adopted which corresponds to $20P$ resulting in providing sufficient space for the highly turbulent flow to pass through downstream region.

For each hydraulic condition as listed in Table 4.1 for both Linear Output Piano Key Weir (LOPKW) and Non-Linear Output Piano Key Weir (NLOPKW), a simulation was first executed for initial flat bed case and then simulation was performed for both LOPKW and NLOPKW for scoured bed profile resulting in total four numerical configurations.

For the equilibrium scour configurations, the experimental bed elevations were set of discrete measurements of scour level and downstream distance. The scour elevations were referenced to the experimental vertical datum, corresponding to z=0 in the numerical domain. The measured profiles were converted from centimetres to metres before being incorporated into the computational geometry.

The experimental hydraulic conditions used to define the numerical models are illustrated schematically in Figures 4.1 and 4.2. The figures show the head over the PKW crest, the weir height, the downstream raised-bed level, and the corresponding downstream water depth for the LOPKW and NLOPKW configurations. These dimensions were used to establish the hydraulic and geometric reference levels for the numerical simulations.

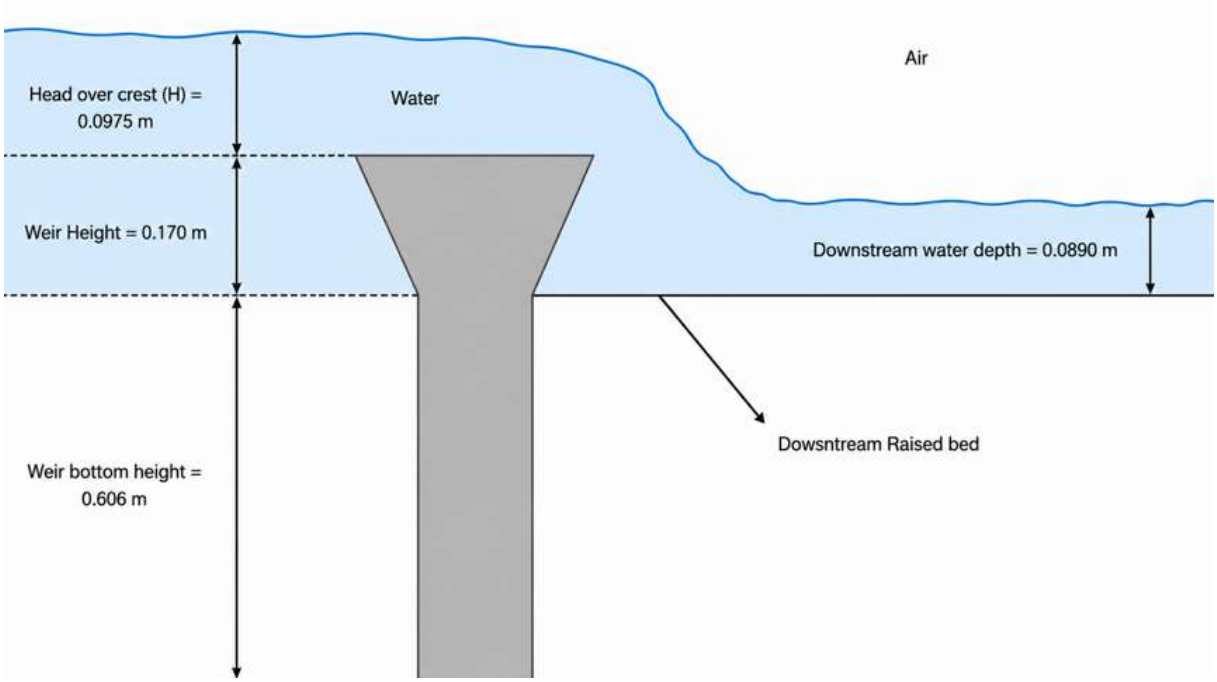


Figure 4.1: Schematic representation of the experimental hydraulic conditions for the LOPKW configuration.

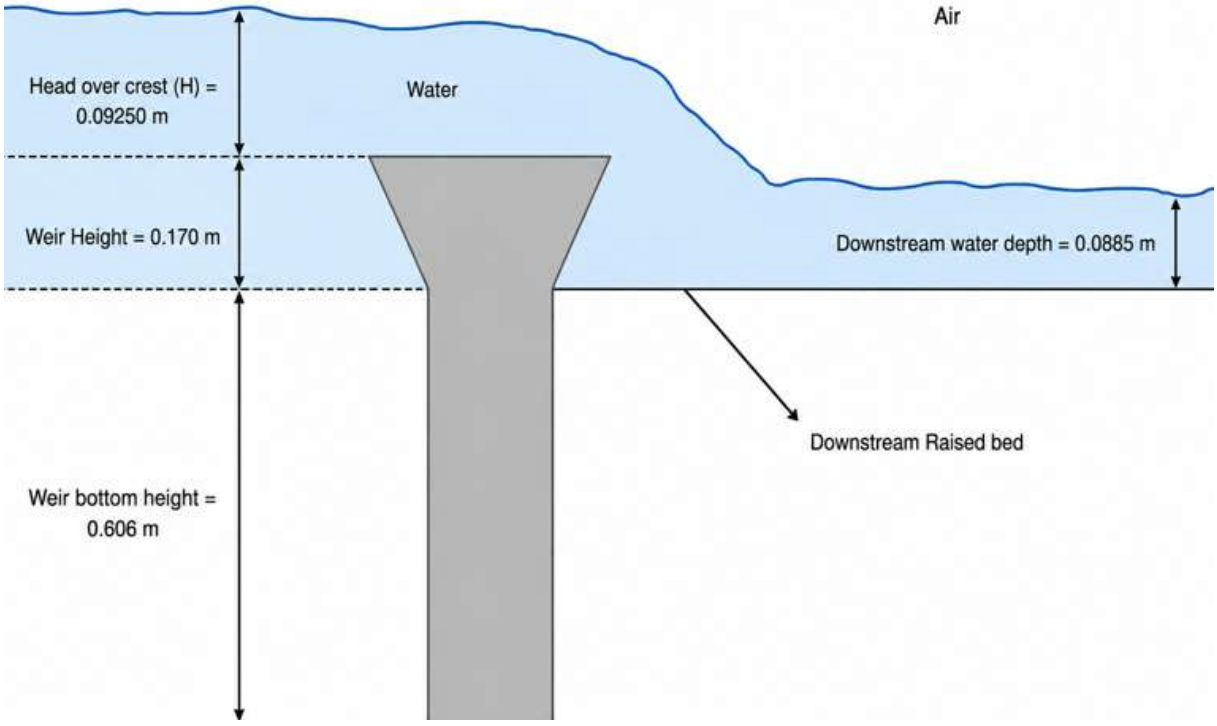


Figure 4.2: Schematic representation of the experimental hydraulic conditions for the NLOPKW configuration.

Table 4.1: Experimental hydraulic conditions considered in the numerical study.

Parameter	**LOPKW**	**NLOPKW**
Discharge, Q (L/s)	110	110
Head (h) over crest (m)	0.0975	0.09250
Upstream water level (m)	0.2965	0.2915
Downstream water level (m)	0.1180	0.1175
Downstream water depth above bed (m)	0.0890	0.0885

4.3 Numerical Case Configurations

Four numerical configurations were developed with same discharge flow rate of $Q = 110$ L/s with the classification of linear and non-linear geometry and by the flat bed and scoured bed case as illustrated in Table 4.2

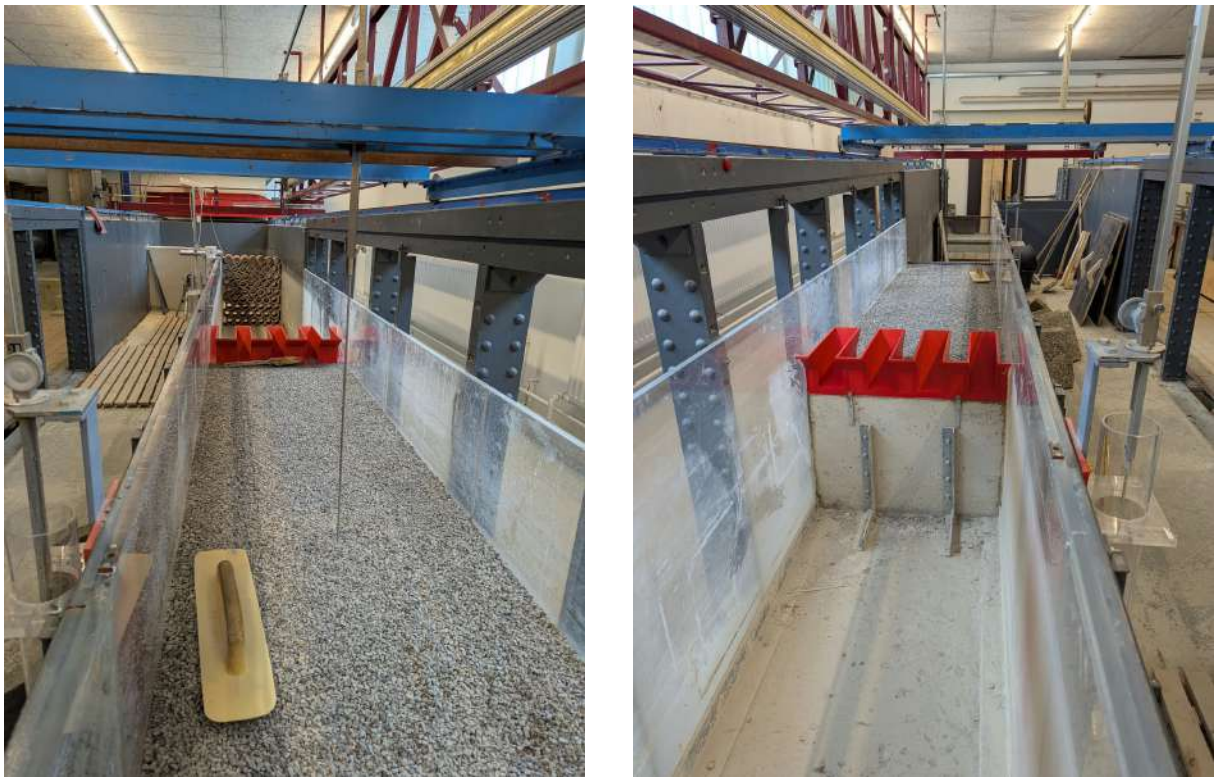
The same general numerical formulation, turbulence model, solver settings were maintained across all cases. This arrangement allows the effects of PKW geometry and downstream bed deformation to be assessed independently while keeping the imposed discharge constant.

Table 4.2: Numerical configurations investigated in the present study.

Case	**Bed configuration**	**Discharge (L/s)**	**PKW Geometry**
Case 1	Flat rigid bed	110	LOPKW
Case 2	Flat rigid bed	110	LOPKW
Case 3	Equilibrium scour bed	110	NLOPKW
Case 4	Equilibrium scour bed	110	NLOPKW

4.3.1 Flat-Bed Configurations

The Flat bed configuration is represented by a fixed horizontal surface at downstream region before the initiation of scour as illustrated in Figure 4.3. The figure presents the experimental flat bed configuration which was used as reference for the numerical model.



(a) Downstream flat-bed configuration.

(b) Flat-bed configuration viewed towards the PKW.

Figure 4.3: Experimental flat-bed configuration used as the reference for the numerical simulations: (a) downstream channel with the horizontal bed and (b) view towards the Piano Key Weir.

Both flat bed simulation for LOPKW and NLOPKW were performed at identical experimental discharge rate of $Q = 110\ \mathrm{L/s}$. The remaining numerical model settings were kept same; the only difference was the geometry difference.

The flat bed cases investigate the velocity gradients adjacent to the bed layer, turbulent kinetic energy, shear stresses formation, vorticity at the corresponding inlet keys, outlet keys and sidewalls of Piano key weirs

4.3.2 Equilibrium Scour-Bed Configurations

Case 3 and Case 4 from Table 4.2 represent the scoured bed configuration which was obtained from the experimental scoured profile of outlet key. The scoured bed configuration was also simulated at the same discharge rate of $Q = 110\ \mathrm{L/s}$.

For the scoured bed configuration, the initial horizontal downstream bed was replaced by the coordinated used by the experimental scoured bed profile. The downstream mesh was generated following the experimental profile. The coordinates used are illustrated in Figure 4.4

The purpose of incorporating the same discharge rate for all cases is to make comparison between flatbed case and scoured bed case. For instance, case 1 can be compared with case 2 whether geometry influences shear stress or not and, we repeat the same process for scoured bed profile. The purpose of different scoured configurations is to investigate how downstream hydraulic changes such as bed shear stress, vorticity and turbulence with changing bed configuration.

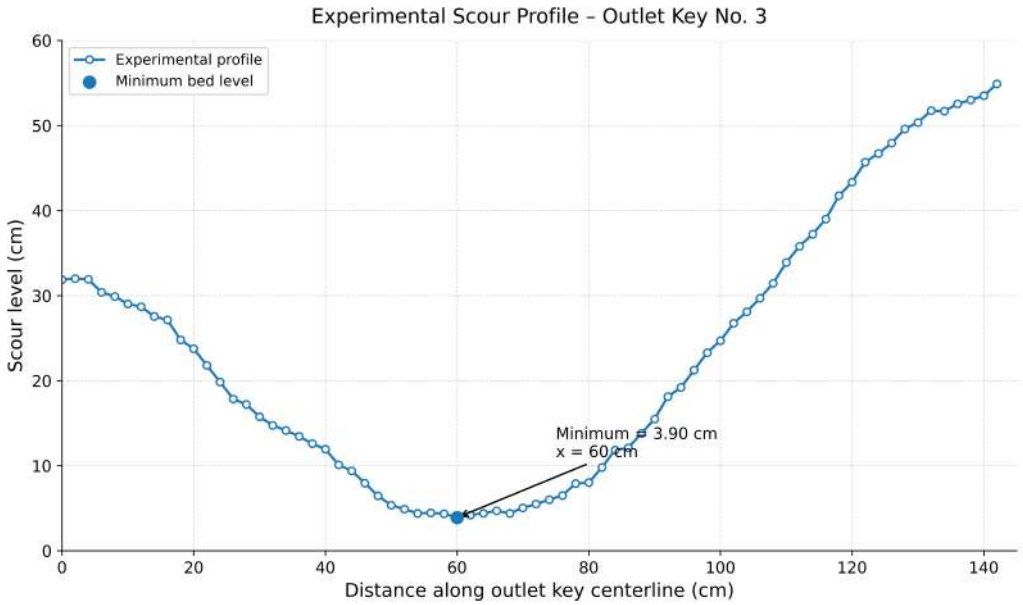


Figure 4.4: Experimental equilibrium scour-bed profile used to construct the numerical downstream bed geometry.

4.4 Computational Domain and Geometry

The computation domain was generated to visualize the experimental channel. The streamwise direction is denoted by x-axis and transverse direction by y-axis, and the vertical direction by the z-axis. An upstream length of 1.50 m was adopted which corresponds to approximately $9P$ to allow the flow to fully develop before it reaches PKW. Downstream length of 3.396 m was adopted which corresponds to $20P$ resulting in providing sufficient space for the highly turbulent flow to pass through downstream region.

The upstream boundary begins at $x = -1.50$ m, while the downstream boundary begins from $x = +0.318$ m which is just after the PKW. The channel width was 0.82 m, and the upper atmospheric region of the computational domain was positioned at $z = 1.05$ m as illustrated in Table 4.3. The downstream raised-bed reference elevation was $z = 0.606$ m.

The Piano key Weir geometry was represented by triangulated surface STL file and was incorporated into blockMesh using `snappyHexMesh`. Two experimental PKW geometries with experimental dimensions are illustrated in Figures 4.5 and 4.6, respectively.

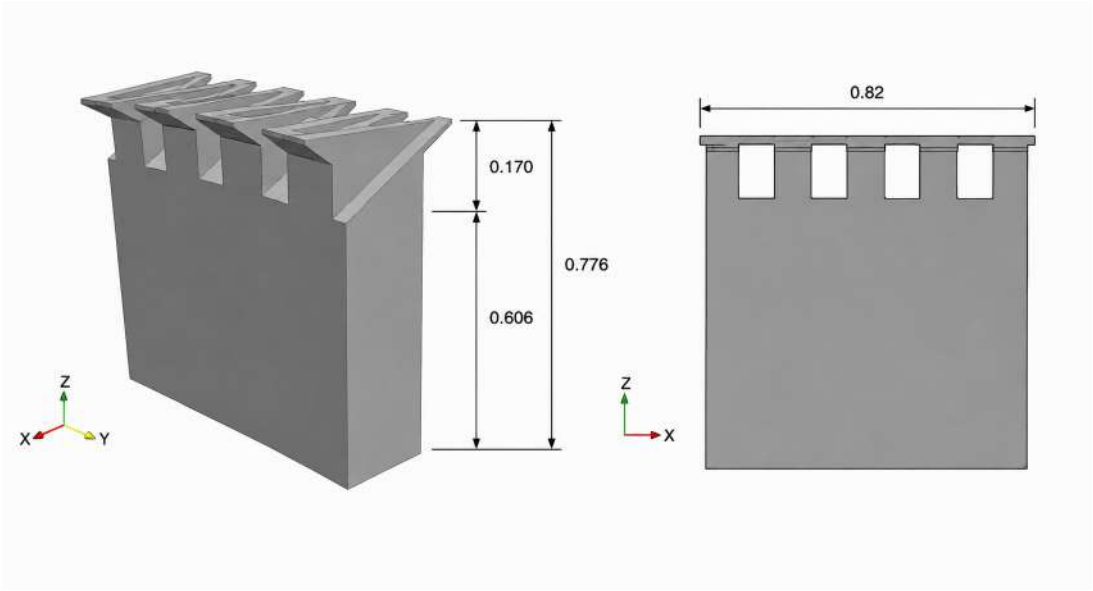


Figure 4.5: Geometry and principal dimensions of the linear Piano Key Weir (LOPKW) configuration used in the numerical simulations.

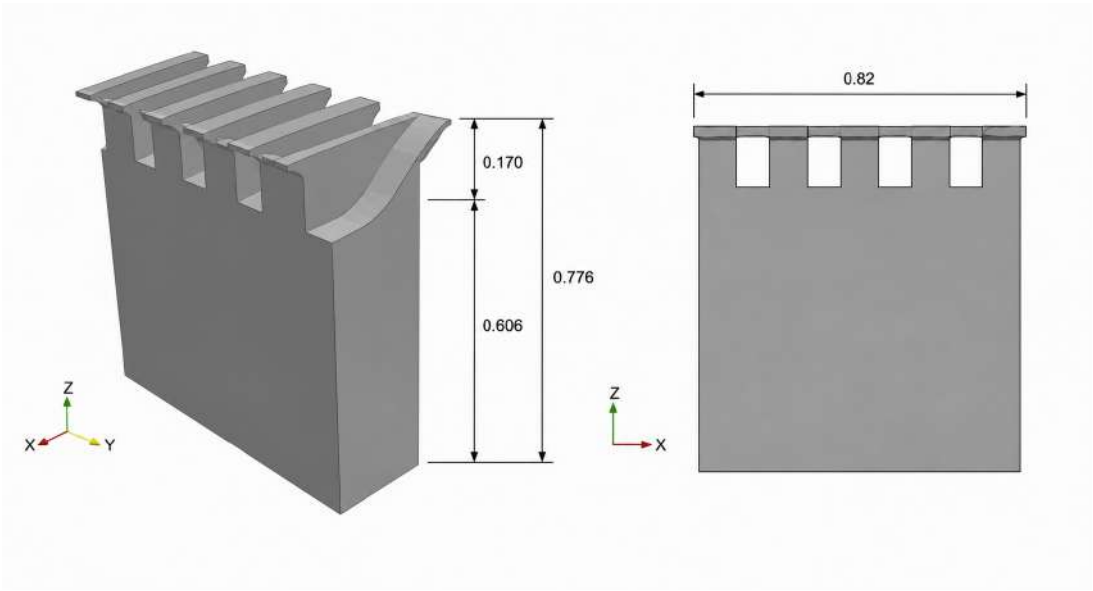


Figure 4.6: Geometry and principal dimensions of the non-linear Piano Key Weir (NLOPKW) configuration used in the numerical simulations.

For the equilibrium scour cases, the downstream bed was modified according to the experimentally measured profile as illustrated in Figure 4.4 while the overall upstream domain, channel width, PKW geometry, and outlet location were kept identical as for Flat bed case. Maintaining a similar computational domain between the flat-bed and scour-bed configurations reduced differences arising from numerical domain selection.

Table 4.3: Principal dimensions of the computational domain.

Parameter	Value (m)
Upstream boundary position	−1.500
Downstream outlet position	3.820
Channel width	0.820
Domain height	1.050
Downstream raised-bed elevation	0.606
PKW-region downstream end	0.318

4.5 Mesh Generation

The computational mesh was generated using a two-stage procedure. First, a structured hexahedral background mesh was generated using `blockMesh`. The background mesh defined the overall computational domain and the principal subdivisions in the streamwise, transverse and vertical directions. In the equilibrium scour configurations, the measured downstream bed profile was also incorporated directly into the `blockMesh` geometry.

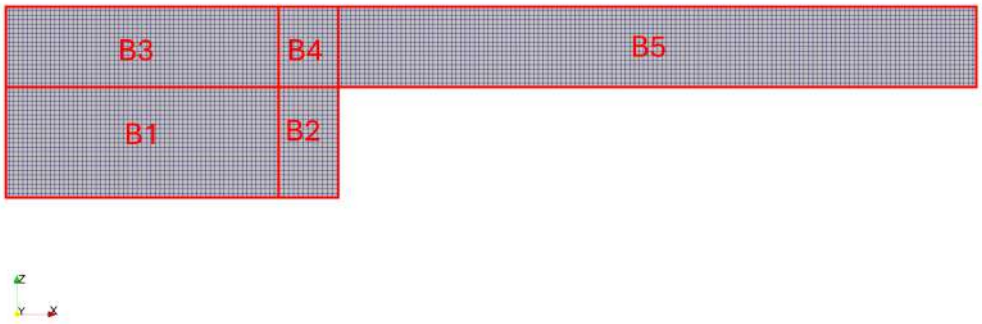
The second stage used `snappyHexMesh` to incorporate the detailed PKW surface geometry and locally refine regions containing strong hydraulic gradients. Refinement was concentrated around the PKW, the crest and nappe region, the expected air-water interface surface region, the downstream jet and recirculation zone, and the equilibrium scour region.

The same general refinement philosophy was maintained between the four cases to allow meaningful comparison of the resulting hydraulic fields.

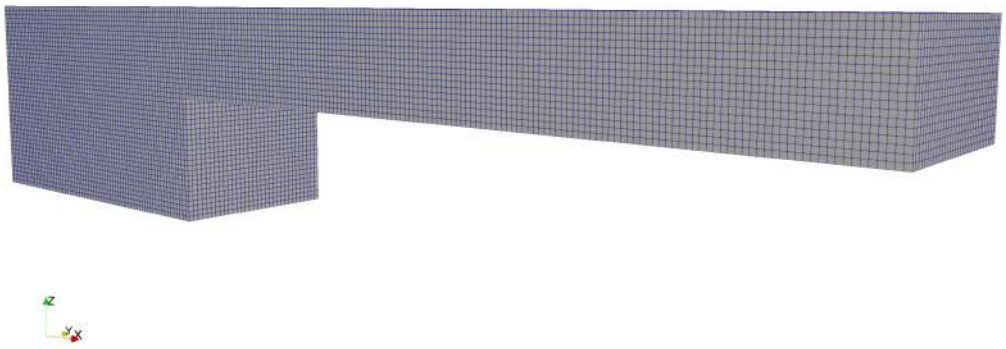
4.5.1 Background Mesh Using `blockMesh`

The initial background computational mesh was generated using the OpenFOAM `blockMesh` utility. The background mesh was divided into five structured hexahedral blocks in order to incorporate required refinement to each block according to flow physics need to be resolved accurately , denoted as B1–B5 in Figure 4.7. Blocks B1 and B3 represent the upstream region, Blocks B2 and B4 cover the PKW region, and Block B5 represents the extended downstream region.

The structured background mesh provides the basic computational domain before the PKW geometry and local refinement are introduced using `snappyHexMesh`. The upstream boundary is located at $x = -1.500$ m, the downstream outlet at $x = 3.820$ m, the channel width is 0.820 m, and the upper boundary is located at $z = 1.050$ m. A base cell size of approximately 25 mm was used for the background `blockMesh` in both the flat-bed and equilibrium scour-bed configurations.



(a) Block decomposition of the computational domain into B1–B5.



(b) Three-dimensional view of the structured background mesh.

Figure 4.7: Background mesh generated using `blockMesh`: (a) block decomposition of the computational domain and (b) three-dimensional representation of the structured mesh.

The resulting background mesh was subsequently used as the base mesh for `snappyHexMesh`, where the PKW geometry and local refinement regions were introduced.

4.5.2 Geometry Refinement Using `snappyHexMesh`

The 3D geometry of Piano key weir was incorporated into snappyHexMesh as an STL file. SurfaceFeatureExtract dictionary was used to extract important surfaces edges from STL file during snapping procedure.

SnappyHexMesh was refined using refinement regions in critical regions to resolve physics more accurately such as near the nappe formation, around PKW surface, near air-water interface

and recirculation region. Overall, the refinement levels were adopted same during all simulation, the only difference was with flatbed case and scoured bed case where in scoured bed, the additional refinement region was added.

The background mesh of cell size of approximately 25 mm was used as initial starting mesh size. In `snappyHexMesh`, each successive refinement level approximately reduce the cell size by half. Therefore, refinement levels 1, 2, and 3 correspond to approximate characteristic cell sizes of 12.5 mm, 6.25 mm, and 3.125 mm, respectively.

Flat-Bed Refinement

For the flat-bed case, the mesh along PKW surface was refined to level 3 corresponding to mesh size of 3.125 mm. The mesh was refined to level 2 (6.25 mm) for the region of PKW volume, Crest interference and nappe region and also the downstream jet and recirculation region was refined to same refinement level. The mesh for the upstream water-air interface region, downstream bulk water region and the outlet region was refined to level 1 (12.5 mm) and downstream water-air interface was refined to level 2(6.25 mm).

The corresponding refinement levels for the corresponding refinement regions are illustrated in Table 4.4 respectively.

Table 4.4: Local refinement strategy used for the flat-bed configurations.

Refinement region	**Refinement level**	**Approx. cell size (mm)**
PKW surface	3	3.125
Upstream interface region	1	12.5
Approach-flow region	1	12.5
PKW volume	2	6.25
Crest and nappe region	2	6.25
Jet and recirculation region	2	6.25
Downstream bulk-water region	1	12.5
Downstream interface region	2	6.25
Outlet region	1	12.5

Equilibrium Scour-Bed Refinement

For the equilibrium scour-bed case, the overall refinement used for the flat bed case was retained in scoured bed case to make a sensible comparison between different numerical configurations. For the scoured bed case, the experimental measured scoured profile was introduced as a separate STL surface named as `scourBed.stl`.

The `scourBed.stl` surface was refined to level 1 corresponding to cell size of 12.5 mm in the scoured region resulting in additional refinement in the scoured bed section. A `scourInteriorBox` refinement box was introduced over the scour bed volume and was refined to level 1.

The jet and recirculation refinement region remained at level 2, corresponding to an approximate characteristic cell size of 6.25 mm, but its lower boundary was maintained at the original

raised-bed elevation, $z = 0.606$ m. Consequently, the deeper portion of the scour cavity was refined using the level 1 scour-interior region rather than extending the level 2 jet refinement throughout the entire scour depth. This approach provided additional resolution in the scour cavity while limiting the generation of unnecessarily small cells and the associated computational cost.

The refinement levels and corresponding approximate characteristic cell sizes used for the equilibrium scour-bed configurations are summarized in Table 4.5.

Table 4.5: Local refinement strategy used for the equilibrium scour-bed configurations.

Refinement region	**Refinement level**	**Approx. cell size (mm)**
PKW surface	3	3.125
Scour-bed surface	1	12.5
Upstream interface region	1	12.5
Approach-flow region	1	12.5
PKW volume	2	6.25
Crest and nappe region	2	6.25
Jet and recirculation region	2	6.25
Scour-interior region	1	12.5
Downstream bulk-water region	1	12.5
Downstream interface region	2	6.25
Outlet region	1	12.5

For both bed configurations, three cells were maintained between successive refinement levels using `nCellsBetweenLevels = 3`, thereby providing a gradual transition between coarse and refined regions. No prism layers were added, and the same snapping and mesh-quality controls were maintained for the flat-bed and scour-bed meshes.

4.5.3 Mesh for Flat-Bed Configurations

The flat-bed meshes were developed to represent the initial downstream geometry prior to scour development. A horizontal rigid bed was maintained downstream of the PKW, and local refinement was applied in the region of jet impact and recirculation.

The same refinement approach was used for both flat-bed operating conditions, with any differences in final cell count arising primarily from the specific case geometry and the castellated refinement process. Particular attention was given to maintaining adequate resolution near the PKW crest, air-water interface surface and downstream bed because these regions contain strong velocity and pressure gradients.

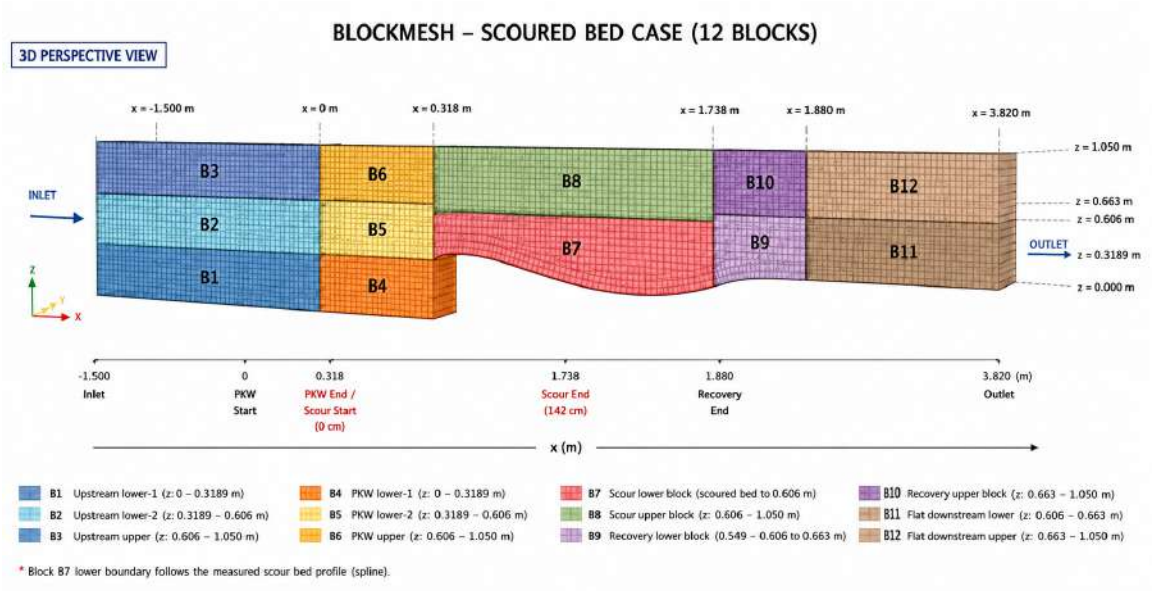
The flat-bed configurations provide the reference mesh against which the corresponding equilibrium scour-bed cases are compared.

4.5.4 Mesh Adaptation for the Equilibrium Scour Profile

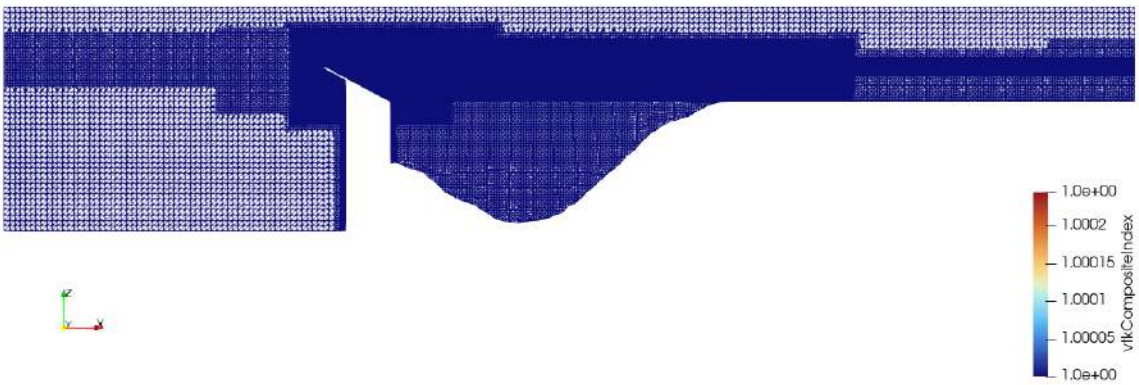
For the equilibrium scour-bed cases, the downstream background mesh was modified to reproduce the measured scour profile. The profile begins at the downstream end of the PKW region and was introduced directly into the `blockMesh` geometry. The downstream domain was divided into additional blocks to allow the lower mesh boundary to follow the curved bed while maintaining a structured hexahedral mesh.

A total of 12 blocks (B1–B12) were used for the scour-bed configuration. The main modification occurs in the downstream region, where Block B7 follows the measured scour profile. Additional blocks were introduced downstream of the scour region to provide a gradual transition from the deformed bed to the horizontal downstream section. The resulting block arrangement and generated background mesh are shown in Figure 4.8.

The adapted background mesh was subsequently used for the `snappyHexMesh` procedure, where the PKW geometry and the required local refinement regions were introduced.



(a) Block arrangement used to represent the equilibrium scour-bed profile (Generated using AI for better Visualization)



(b) Generated structured background mesh following the scour-bed profile.

Figure 4.8: Background mesh for the equilibrium scour-bed configuration: (a) decomposition of the computational domain into 12 blocks and (b) resulting structured mesh adapted to the measured scour profile.

4.6 Governing Equations

The numerical simulations were executed using `interFoam` which is commonly used for transient flows for incompressible two immiscible phases such as water and air and the interface between two phases is resolved using multiphase model called as Volume of Fluid (VOF) method. The governing equations comprise of conservation of mass, momentum and transport equations for volume fraction for water. Turbulence effects are modelled using Reynolds Averaged Navier Stokes (RANS) together with k–ω Shear Stress Transport (SST) turbulence model.

4.6.1 Continuity Equation

For an incompressible flow, conservation of mass requires the velocity field to satisfy the continuity equation [20],

$$\nabla \cdot \mathbf{U} = 0, \tag{4.1}$$

where $\mathbf{U}$ is the fluid velocity vector. Equation 4.1 ensures conservation of mass throughout the computational domain.

4.6.2 Momentum Equation

The motion of the air–water mixture is governed by the incompressible Reynolds-Averaged Navier–Stokes momentum equation. In OpenFOAM, the pressure treatment for flows subjected to gravity is formulated using the modified pressure field p_{rgh} [21]. In general form, the momentum equation can be expressed as

$$\frac{\partial(\rho\mathbf{U})}{\partial t} + \nabla \cdot (\rho\mathbf{U}\mathbf{U}) = -\nabla p + \nabla \cdot \left[\mu_{\text{eff}} \left(\nabla\mathbf{U} + \nabla\mathbf{U}^T\right)\right] + \rho\mathbf{g} + \mathbf{F}_\sigma, \tag{4.2}$$

where ρ is the local mixture density, $\mathbf{U}$ is the velocity vector, p is the pressure, $\mathbf{g}$ is the gravitational acceleration, and $\mathbf{F}_\sigma$ represents the surface-tension force acting at the air–water interface. The effective dynamic viscosity is expressed as

$$\mu_{\text{eff}} = \mu + \mu_t, \tag{4.3}$$

where μ is the molecular dynamic viscosity and μ_t is the turbulent dynamic viscosity obtained from the turbulence model.

In OpenFOAM, the pressure treatment for flows subjected to gravity is formulated using the modified pressure field p_{rgh}, which removes the hydrostatic contribution from the pressure field. It is related to the physical pressure through

$$p_{rgh} = p - \rho\,\mathbf{g} \cdot \mathbf{h}, \tag{4.4}$$

where $\mathbf{h}$ represents the position vector relative to the gravitational reference. This formulation is particularly convenient for free-surface flows because the hydrostatic contribution is separated from the pressure field solved numerically.

4.6.3 Volume of Fluid Method

The air-water interface surface between water and air was captured using the Volume of Fluid (VOF) method. The VOF method represents the different phases through a scalar phase-fraction variable and is widely used for the numerical simulation of free-surfac flows [22].

In the present simulations, the water volume fraction is denoted by α, corresponding to the OpenFOAM field `alpha.water`. Its value is defined as

$$\alpha = \begin{cases} 1, & \text{water}, \\ 0, & \text{air}, \\ 0 < \alpha < 1, & \text{air–water interface}. \end{cases} \tag{4.5}$$

Therefore, a computational cell with $\alpha = 1$ is completely occupied by water, while $\alpha = 0$ represents a cell containing only air. Intermediate values indicate cells through which the air–water interface passes.

The transport of the water volume fraction is represented by

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{U}) + \nabla \cdot [\alpha(1-\alpha)\mathbf{U}_r] = 0, \tag{4.6}$$

where $\mathbf{U}_r$ is an artificial relative compression velocity used to maintain a sharp numerical interface between water and air. The additional compression term acts primarily within interface cells because $\alpha(1-\alpha)$ approaches zero in cells containing only one phase.

Mixture Properties

Because a single momentum equation is solved for the air–water mixture, the local fluid properties depend on the water volume fraction. The mixture density is determined as

$$\rho = \alpha \rho_w + (1-\alpha)\rho_a, \tag{4.7}$$

where ρ_w and ρ_a are the densities of water and air, respectively.

Similarly, the mixture dynamic viscosity is obtained from

$$\mu = \alpha \mu_w + (1-\alpha)\mu_a, \tag{4.8}$$

where μ_w and μ_a are the dynamic viscosities of water and air, respectively. Consequently, the fluid properties vary continuously across the numerically captured air–water interface.

The VOF formulation is particularly suitable for the present application because the flow over the Piano Key Weir involves a strongly deforming air-water interface, nappe formation, jet impact and downstream interface surface disturbances.

4.7 Turbulence Modelling

The Flow over the Piano Key Weir is determined by velocity gradients, Jet formation, recirculation zone, crest, shear tresses, directly resolving all the turbulent scales same as in Direct Numerical Simulation (DNS) is extremely expensive and requires massive computational power. Therefore, the RANS approach models the Reynold stresses through turbulence models without resolving turbulence fluctuations at all scales.

4.7.1 k–ω SST Model

The Reynolds-Averaged Navier–Stokes equations were closed using the k–ω Shear Stress Transport (SST) turbulence model. The SST formulation combines the near-wall behaviour of the k–ω model with a transformed k–ε behaviour away from solid boundaries through blending functions. This makes the model suitable for flows involving strong adverse pressure gradients and flow separation [23, 24].

The model solves two additional transport equations: one for the turbulent kinetic energy k and another for the specific dissipation rate ω. The transport equation for turbulent kinetic energy can be written in general form as

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho \mathbf{U} k) = P_k - \beta^* \rho k \omega + \nabla \cdot [(\mu + \sigma_k \mu_t) \nabla k] \,, \tag{4.9}$$

where P_k represents the production of turbulent kinetic energy, β^* is a model coefficient, μ_t is the turbulent dynamic viscosity, and σ_k controls the diffusion of turbulent kinetic energy.

The transport equation for the specific dissipation rate ω is expressed in general form as

$$\frac{\partial(\rho \omega)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \omega) = \frac{\gamma \omega}{k} P_k - \beta \rho \omega^2 + \nabla \cdot [(\mu + \sigma_\omega \mu_t) \nabla \omega] + D_\omega, \tag{4.10}$$

where γ, β, and σ_ω are model coefficients and D_ω represents the cross-diffusion contribution associated with the SST formulation.

The turbulent eddy viscosity is subsequently determined from the local values of k and ω, with the SST formulation applying a shear-stress limiter. In general, it can be expressed as

$$\mu_t = \rho \frac{a_1 k}{\max(a_1 \omega,\ S F_2)}, \tag{4.11}$$

where a_1 is an SST model coefficient, S represents the characteristic strain-rate magnitude, and F_2 is one of the SST blending functions [23, 24].

The k–ω SST model was applied consistently to all numerical configurations investigated in the present study. Consequently, differences between the simulated cases arise from changes in the PKW geometry and downstream bed configuration rather than from changes in the turbulence modelling approach.

4.8 Wall Shear Stress Evaluation

Wall shear stress was evaluated to quantify the tangential hydraulic loading acting on the solid boundaries, with particular attention given to the downstream bed. This quantity is important for the present study because regions of elevated bed shear stress indicate locations where the hydraulic forces acting on the bed are comparatively high.

In OpenFOAM, the wall shear-stress field is obtained from the near-wall stress tensor. The wall shear-stress vector represents the tangential component of the stress acting at a solid boundary and may be expressed in general form as

$$\boldsymbol{\tau}_w = \mathbf{n} \cdot \boldsymbol{\tau} - [(\mathbf{n} \cdot \boldsymbol{\tau}) \cdot \mathbf{n}]\, \mathbf{n}, \tag{4.12}$$

where $\mathbf{n}$ is the unit vector normal to the wall and $\boldsymbol{\tau}$ represents the effective stress tensor.

For a Reynolds-Averaged Navier–Stokes calculation, the near-wall stress contains contributions from both the molecular and turbulent viscosity. The effective viscosity is therefore expressed as

$$\mu_{\text{eff}} = \mu + \mu_t, \tag{4.13}$$

where μ is the molecular dynamic viscosity and μ_t is the turbulent dynamic viscosity predicted by the turbulence model.

The magnitude of wall shear stress may be related conceptually to the tangential velocity gradient at the wall through

$$\tau_w = \mu_{\text{eff}} \left. \frac{\partial U_t}{\partial n} \right|_w, \tag{4.14}$$

where U_t denotes the velocity component tangential to the wall and n represents the wall-normal direction.

In the present simulations, the OpenFOAM `wallShearStress` function object was used to calculate the wall shear-stress field on the downstream bed. The resulting vector field was subsequently processed in ParaView, where its magnitude was evaluated and visualised using surface contours and line-based sampling.

When the `wallShearStress` output is expressed as a kinematic stress, with dimensions of $\mathrm{m}^2/\mathrm{s}^2$, conversion to dimensional shear stress is required before comparison with the critical sediment shear stress. The corresponding dimensional bed shear stress is obtained from

$$\tau_b = \rho_w \left| \boldsymbol{\tau}_{w,\mathrm{OF}} \right|, \tag{4.15}$$

where ρ_w is the density of water and $\boldsymbol{\tau}_{w,\mathrm{OF}}$ is the wall shear-stress vector obtained from OpenFOAM. The resulting value of τ_b is expressed in $\mathrm{N/m^2}$ (Pa) and can therefore be compared directly with the critical shear stress of the sediment.

4.9 Critical Shear Stress for flat bed for Sediment Motion

The critical bed shear stress represents the threshold hydraulic loading at which sediment particles begin to move. The transition from a stationary bed to the initial movement of sediment particles is commonly referred to as the condition of incipient motion [25]. The prediction of this threshold is generally based on the Shields approach, in which the hydrodynamic force acting on the sediment is related to the submerged weight of the particles.

The dimensionless Shields parameter is defined as

$$\theta = \frac{\tau_b}{(\rho_s - \rho_w) g\, d}, \tag{4.16}$$

where τ_b is the applied bed shear stress, ρ_s is the sediment density, ρ_w is the water density, g is gravitational acceleration, and d is the representative sediment diameter.

At the threshold of sediment motion, Equation 4.16 becomes

$$\theta_{cr} = \frac{\tau_{cr}}{(\rho_s - \rho_w) g\, d}, \tag{4.17}$$

where θ_{cr} is the critical Shields parameter and τ_{cr} is the corresponding critical bed shear stress.

Paphitis [25] developed analytical expressions for estimating sediment threshold conditions from an extensive compilation of experimental data. In the present study, the mean threshold relationship based on the dimensionless grain diameter was adopted. The critical Shields parameter is calculated as

$$\theta_{cr} = \frac{0.273}{1 + 1.2 D_*} + 0.0460 \left(1 - 0.576 \exp(-0.02 D_*)\right), \tag{4.18}$$

where D_* is the dimensionless grain diameter. This relationship corresponds to the mean threshold curve proposed by Paphitis [25].

The dimensionless grain diameter is determined from

$$D_* = d \left[\frac{(s-1) g}{\nu^2} \right]^{1/3}, \tag{4.19}$$

where s is the relative sediment density,

$$s = \frac{\rho_s}{\rho_w}, \tag{4.20}$$

and ν is the kinematic viscosity of water.

For the sediment used in the experimental investigation, the representative grain diameter was

$$d = 6.5 \text{ mm} = 6.5 \times 10^{-3} \text{ m}. \tag{4.21}$$

Using the corresponding sediment and water properties in Equation 4.19, the mean threshold relationship in Equation 4.18 gives a critical Shields parameter of approximately

$$\theta_{cr} \approx 0.0464. \tag{4.22}$$

Once the critical Shields parameter is known, the dimensional critical bed shear stress is obtained by rearranging Equation 4.17:

$$\tau_{cr} = \theta_{cr}(\rho_s - \rho_w) g\, d. \tag{4.23}$$

Using the relative density $s = \rho_s/\rho_w$, the same expression can be written as

$$\tau_{cr} = \theta_{cr}\rho_w(s - 1) g\, d. \tag{4.24}$$

For the 6.5 mm sediment considered in the present study, this procedure resulted in

$$\boxed{\tau_{cr} \approx 4.9 \text{ N/m}^2} \tag{4.25}$$

which was adopted as the reference critical shear stress for the comparison with the numerical bed shear-stress results.

Since τ_{cr} is determined from the sediment properties and fluid properties rather than directly from the imposed discharge, the same critical shear-stress value was used for all numerical configurations in the present investigation. Locations satisfying

$$\tau_b > \tau_{cr} \tag{4.26}$$

were interpreted as regions where the computed hydraulic loading exceeds the estimated mean threshold for sediment motion.

It should be noted that the threshold of sediment motion is not an exact deterministic quantity. Experimental threshold measurements exhibit scatter because sediment entrainment is influenced by factors such as particle packing, exposure, shape, turbulence and the adopted definition of incipient motion [25]. Consequently, the value of $\tau_{cr} = 4.9$ N/m^2 is treated in the present study as a mean engineering threshold rather than an absolute boundary between sediment motion and no motion.

4.10 Boundary and Initial Conditions

Appropriate boundaries and initial conditions were applied after consulting previous numerical studies in literature to achieve experimentally hydraulic conditions naturally within computation domain. Same boundary conditions were replicated to all numerical cases except the modification of PKW geometry and bed geometry.

At the inlet, constant experimental discharge was prescribed using the discharge of $Q = 0.110\ \mathrm{m}^3/\mathrm{s}$ (110 L/s) `variableHeightFlowRateInletVelocity` boundary condition. This condition allows to maintain constant discharge flow rate of (110 L/s) while adjusting the inlet velocity according to water level as water level continuously fluctuates during the simulation. The inlet pressure field p_{rgh} was assigned the `fixedFluxPressure` condition so that the pressure could adjust consistently with the imposed velocity flux.

At the downstream outlet, the velocity field was prescribed using the `pressureInlet OutletVelocity` condition. This boundary condition enables the flow to exit the computational domain naturally while also providing appropriate treatment in the event of local reverse flow. The reduced pressure p_{rgh} was fixed to zero at the outlet, thereby providing the pressure reference for the computational domain. The water volume fraction at the outlet was assigned an `inletOutlet` condition, with an inlet value of $\alpha_{\text{water}} = 0$ in the event of reverse flow.

The upper or top boundary of computational domain was maintained as an open atmospheric boundary with a `pressureInletOutletVelocity` condition for the velocity fields and `prghTotalPressure` for the pressure with a reference or initial value of zero. The phase fraction was treated with an `inletOutlet` condition with $\alpha_{\text{water}} = 0$ for incoming flow, representing air entering through the atmospheric boundary.

No slip boundary conditions for the velocity fields were imposed to all solid patches or surfaces including lower PKW bottom, downstream bed, upstream bottom, channel side walls and `fixedFluxPressure` condition was imposed for p_{rgh} and `zeroGradient` was imposed for water phase fraction.

For the turbulent kinetic energy k, a fixed inlet value of $8.95 \times 10^{-5}\ \mathrm{m}^2/\mathrm{s}^2$ was imposed. At the outlet and atmospheric boundaries, an `inletOutlet` condition was used, while the solid boundaries employed the `kqRWallFunction` wall treatment.

The initial velocity field was set to zero throughout the computational domain,

$$\mathbf{U}_0 = (0, 0, 0), \tag{4.27}$$

and the initial reduced pressure was prescribed as

$$p_{rgh,0} = 0. \tag{4.28}$$

The initial water level distribution of the water and air phases was generated using the OpenFOAM `setFields` utility. The default phase fraction was first set to

$$\alpha_{\text{water}} = 0, \tag{4.29}$$

corresponding to air. Selected regions were subsequently initialized with

$$\alpha_{\text{water}} = 1, \tag{4.30}$$

corresponding to water.

For the upstream and PKW region, water was initialized through set Fields utility from $x = -1.500$ m to $x = 0.424$ m up to water height of $z = 0.700$ m which just below the crest of weir. Downstream water height was initialized from the raised bed beginning at $z = 0.606$ m and extending up to water height of $z = 0.690$ m which is 5 cm below the experimental height $z = 0.695$ m. The height of water from the raised bed during initialization phase comes out to be 0.084 m. The remaining portion of the domain was initialized as air.

4.11 Numerical Schemes and Solver Settings

The transient multiphase simulations were performed using the `interFoam` solver in OpenFOAM v2412. The solver employs a finite-volume formulation for the solution of the momentum, pressure, and phase-fraction equations. Pressure–velocity coupling was handled through the PIMPLE framework. In the present simulations, one outer corrector, three pressure correctors, and one non-orthogonal corrector were employed during each time step.

Temporal discretization was performed using a first-order Euler scheme. Spatial gradients were evaluated using a Gauss linear formulation. The momentum convection term was discretized using the `Gauss limitedLinearV 1` scheme, while the volume-fraction transport equation employed the `Gauss vanLeer` scheme. Upwind discretization was used for the transport of the turbulent kinetic energy k and specific dissipation rate ω. Laplacian terms were discretized using the `Gauss linear corrected` scheme, while corrected surface-normal gradients were employed.

The phase fraction was solved using the MULES procedure. One phase-fraction correction and two alpha sub-cycles were employed, with an interface compression coefficient of $c_\alpha = 1$. The interface-compression term was discretized using a Gauss linear scheme. These settings were used to limit numerical diffusion and maintain a relatively sharp air–water interface.

An adaptive time-step strategy was used during the simulations. The time step was automatically adjusted according to prescribed Courant-number limits, allowing smaller time steps during periods of rapid flow acceleration or interface motion while permitting larger values when the solution remained numerically stable.

The same principal numerical schemes were maintained between all four cases to ensure consistency in the comparison of the resulting hydraulic fields.

Table 4.6: Principal numerical discretization schemes used in the simulations.

Term	Numerical scheme
Time derivative	Euler
Gradient terms	Gauss linear
Momentum convection	Gauss limitedLinearV 1
Phase-fraction convection	Gauss vanLeer
Interface compression	Gauss linear ($c_\alpha = 1$)
Turbulent kinetic energy convection	Gauss upwind
Specific dissipation rate convection	Gauss upwind
Laplacian terms	Gauss linear corrected
Surface-normal gradient	corrected

4.12 Simulation Procedure and Post-Processing

After generation of mesh, initial and boundary condition were prescribed to each numerical case and the initial water column heights were initialized using `setFields` utility. Following the completion of boundary and initial conditions, the computational domain was decomposed into multiple processors for parallel and fast execution on high-performance clusters resulting in less computation time.

The transient simulations were executed using the `interFoam` solver in OpenFOAM v2412 on the CoolMUC-4 high-performance computing cluster operated by the Leibniz Supercomputing Centre (LRZ), Germany. The computational mesh and domain were decomposed into 64 processors for parallel processing on the `cm4_inter` partition for a maximum wall-clock duration of eight hours.

An adaptive time-step procedure was adopted during the complete transient simulation. The initial time-step size was specified as $\Delta t = 1 \times 10^{-4}$ s and time step was adjusted automatically during the simulation to satisfy the maximum courant number restriction 0.75 while the maximum allowable time limit was restricted to 1×10^{-2} s.

Computational fields were written at write interval of 5 s of simulation time. These saved solution fields were subsequently used for temporal and spatial post-processing of the flow.

Several function objects were introduced in the `controlDict` to monitor the hydraulic quantities such as inlet flow rate, Outlet flow rate, total water volume and upstream water level within the computational domain to verify whether the flow has statistically reached quasi-steady state or not.

At each 100 solver time steps, the hydraulic quantities such as inlet flow rate, outlet flow rate, total water volume and upstream water level were recorded during simulation and data file for each quantity was saved in separate post processing folders and to visualize these results. Pythons' scripts were created and results were plotted in python. The simulation was stable and steady when these monitored quantities no longer fluctuate or drift and instead, they depict constant or fluctuating trend around stable mean values.

The post processing Analysis for the hydraulic quantities were performed using Para View.

The air-water interface was visualized primarily by using water phase fraction of $\alpha_{\text{water}} = 0.5$. Velocity magnitude, Turbulent kinetic energy and bed shear stress were analyzed and visualized using contours, clips, slices, streamlines and plot over lines to investigate the 3d flow structure along upstream and downstream channel.

Attention was given to the distribution of bed shear stresses along the raised bed and scour bed. The `wallShearStress` field variable was evaluated at the downstream bed, inlet ket, outlet key and sidewalls and raised bed using clips, slices, streamlines, plot over lines to highlight spatial variation of shear stress subjected to elevated hydraulic loading. The results were then compared between flat bed case and scoured bed case with the variation of LOPKW and NLOPKW geometries to investigate how changing bed configurations and PKW geometries effects the near bed flow dynamics and associated bed shear stress.

The four computational cases were analyzed using the same general post-processing methodology so that differences in velocity magnitude, air-water interface behaviour, turbulent flow characteristics, and bed shear stress could be compared consistently. Wherever direct comparisons were performed, equivalent post-processing locations and analysis procedures were maintained between the different configurations.

Chapter 5

Results and Discussion

This chapter presents and discusses the numerical results obtained for the different Piano Key Weir (PKW) configurations investigated in this study. The analysis focuses on the establishment of statistically quasi-steady flow conditions, validation of the numerical model, mean-flow characteristics, turbulence behaviour, streamline patterns, and bed shear-stress distributions. Particular attention is given to the effects of PKW geometry and downstream bed configuration on the hydraulic behaviour.

Four numerical configurations were considered by combining linear and nonlinear PKW geometries with flat-bed and equilibrium scour-bed conditions. The configurations are summarized in Table 5.1.

Table 5.1: Numerical configurations considered in the results analysis.

Case	PKW geometry	Bed configuration
Case 1	Linear	Flat bed
Case 2	Nonlinear	Flat bed
Case 3	Linear	Equilibrium scour bed
Case 4	Nonlinear	Equilibrium scour bed

5.1 Statistically Quasi-Steady Flow Assessment

Before analysing the time-averaged flow fields, the temporal behaviour of the numerical solutions was examined to determine whether statistically quasi-steady hydraulic conditions had been established. The assessment was based on four monitored quantities: inlet and outlet water discharge, total water volume within the computational domain, upstream water level, and the relative difference between inlet and outlet discharge.

The initial part of each simulation was characterized by a transient adjustment of the air-water interface surface and downstream flow. Following this initial period, the inlet and outlet discharges approached one another, while the upstream water level became approximately constant.

The total water volume also ceased to exhibit substantial long-term variation. These observations indicate that the dominant hydraulic quantities had reached statistically stable conditions.

For the final statistical analysis, the interval from $t = 60$ s to $t = 70$ s was selected as the averaging window. This interval is highlighted in the steady-state assessment plots. Although small instantaneous fluctuations remain because the simulations are transient RANS–VOF calculations, the absence of significant long-term drift during the averaging period allows time-averaged quantities to be used for the subsequent hydraulic comparisons.

5.1.1 Flat-Bed Linear PKW

Figure 5.1 shows the temporal evolution of the monitored quantities for the linear PKW with the flat downstream bed. After the initial transient stage, the outlet discharge approaches the imposed inlet discharge, the upstream air-water interface surface elevation becomes nearly constant, and the total water volume stabilizes. The inlet–outlet discharge difference also approaches approximately zero during the selected averaging period.

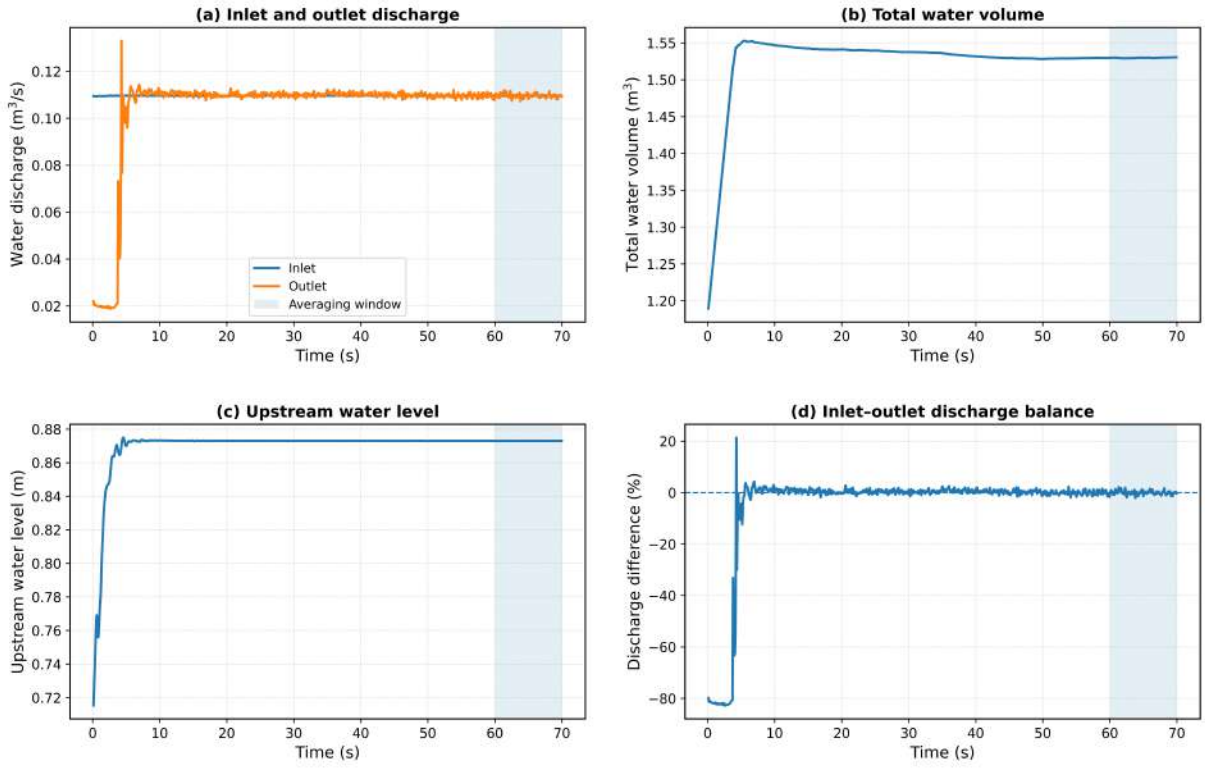


Figure 5.1: Assessment of statistically quasi-steady conditions for the flat-bed linear PKW configuration: (a) inlet and outlet discharge, (b) total water volume, (c) upstream water level, and (d) inlet–outlet discharge balance. The shaded region represents the selected averaging interval from 60 to 70 s.

5.1.2 Flat-Bed Nonlinear PKW

A similar temporal behaviour is observed for the nonlinear flat-bed configuration, as shown in Figure 5.2. The initial transient response is followed by a statistically stable period in which

the monitored hydraulic quantities fluctuate around approximately constant mean values. The interval from 60 to 70 s was therefore also adopted for the calculation of time-averaged fields in this case.

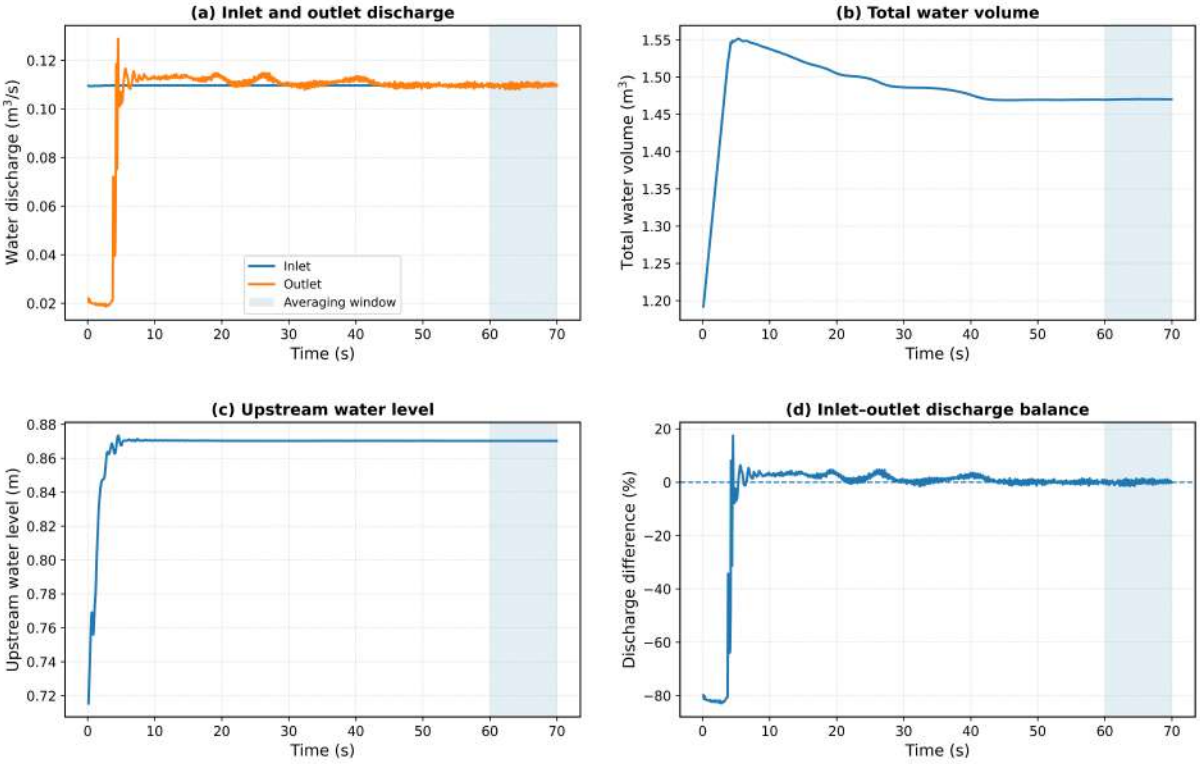


Figure 5.2: Assessment of statistically quasi-steady conditions for the flat-bed nonlinear PKW configuration. The shaded region denotes the averaging interval from 60 to 70 s.

5.1.3 Equilibrium Scour-Bed Linear PKW

The corresponding assessment for the linear PKW with the equilibrium scour-bed geometry is presented in Figure 5.3. The presence of the deeper downstream scour cavity modifies the total water volume and produces a different transient adjustment compared with the flat-bed configuration. Nevertheless, the monitored quantities become statistically stable before the selected averaging interval.

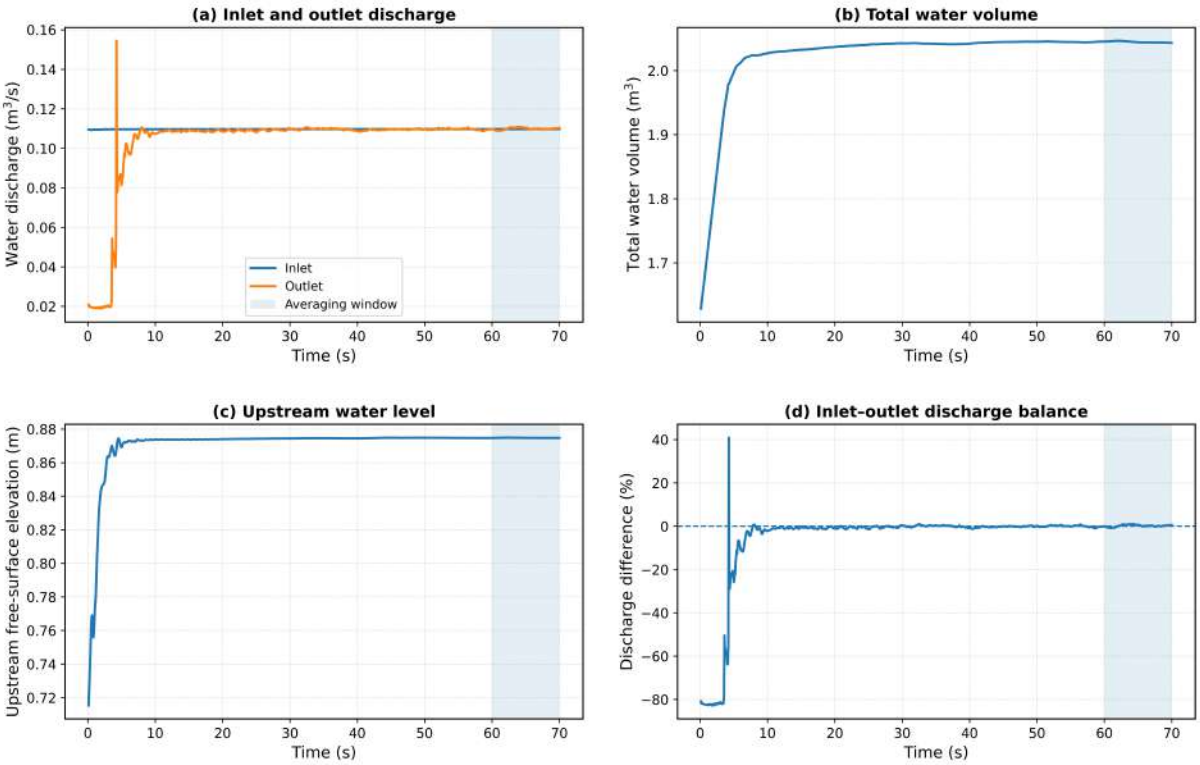


Figure 5.3: Assessment of statistically quasi-steady conditions for the equilibrium scour-bed linear PKW configuration. The shaded region represents the averaging interval from 60 to 70 s.

5.1.4 Validation Against Experimental Data

The numerical model was validated against the experimental hydraulic data using the discharge coefficient of the Piano Key Weir. The discharge coefficient was calculated from the discharge rate at the inlet, total developed crest length, gravitational acceleration, and head over the crest using

$$C_e = \frac{Q}{\frac{2}{3} L \sqrt{2g}\, H^{3/2}}, \tag{5.1}$$

where Q is the discharge rate at the inlet, L is the total developed crest length, g is the gravitational acceleration, and H is the head over the crest. The same formulation was used for both the experimental and numerical results to provide a consistent comparison. The parameters used for the validation were $Q = 0.110\ \mathrm{m}^3/\mathrm{s}$, $L = 4.212\ \mathrm{m}$, and $g = 9.81\ \mathrm{m/s}^2$.

For the LOPKW configuration, the experimental head over the crest was $H_{\mathrm{exp}} = 0.0975$ m, resulting in a discharge coefficient of $C_{e,\mathrm{exp}} = 0.2905$. For the flat-bed numerical case, the corresponding CFD head was $H_{\mathrm{CFD}} = 0.0930$ m, giving $C_{e,\mathrm{CFD}} = 0.3118$. The relative difference between the numerical and experimental discharge coefficients was therefore 7.35%, while the difference in head over the crest was approximately 4.62%.

For the scoured-bed LOPKW configuration, the CFD head over the crest was $H_{\mathrm{CFD}} = 0.0970$ m, giving a discharge coefficient of $C_{e,\mathrm{CFD}} = 0.2927$. The corresponding difference

from the experimental discharge coefficient was approximately 0.77%, while the difference in head over the crest was approximately 0.51%.

For the NLOPKW configuration, the experimental head over the crest was $H_{\text{exp}} = 0.0925$ m, corresponding to $C_{e,\text{exp}} = 0.3144$. For the flat-bed numerical case, the CFD prediction gave $H_{\text{CFD}} = 0.0900$ m and $C_{e,\text{CFD}} = 0.3276$. The resulting relative difference in discharge coefficient was 4.20%, while the head difference was approximately 2.70%.

For the scoured-bed NLOPKW configuration, the CFD head over the crest was $H_{\text{CFD}} = 0.0930$ m, resulting in a discharge coefficient of $C_{e,\text{CFD}} = 0.3118$. The difference from the experimental discharge coefficient was approximately 0.81%, while the difference in head over the crest was approximately 0.54%.

Table 5.2: Comparison between experimental and CFD discharge coefficients for the flat-bed and scoured-bed configurations.

Bed condition	**Configuration**	H_{exp} (m)	H_{CFD} (m)	$C_{e,\text{exp}}$ (-)	$C_{e,\text{CFD}}$ (-)	ΔC_e (%)	ΔH (%)
Flat bed	LOPKW	0.0975	0.0930	0.2905	0.3118	7.35	4.62
Flat bed	NLOPKW	0.0925	0.0900	0.3144	0.3276	4.20	2.70
Scoured bed	LOPKW	0.0975	0.0970	0.2905	0.2927	0.77	0.51
Scoured bed	NLOPKW	0.0925	0.0930	0.3144	0.3118	0.81	0.54

The results show good agreement between the CFD predictions and the experimental measurements for both PKW configurations. For the flat-bed cases, the maximum difference in discharge coefficient was 7.35% for the LOPKW, while the NLOPKW showed a smaller difference of 4.20%.

For the scoured-bed cases, the agreement was even closer. The difference in discharge coefficient was approximately 0.77% for the LOPKW and 0.81% for the NLOPKW. Similarly, the differences in head over the crest were approximately 0.51% and 0.54%, respectively which are illustrated in Table 5.2. These results indicate that the numerical model reproduced the upstream hydraulic performance of both PKW configurations with good accuracy for both the flat-bed and equilibrium scoured-bed conditions.

It should be noted that the agreement in discharge coefficient does not imply equally good agreement in all downstream hydraulic quantities. In particular, the numerical downstream water levels showed larger differences from the experimental values. This discrepancy is discussed separately because the experimental downstream bed slope was not available and the downstream flow was allowed to develop naturally without prescribing the measured downstream water level. Consequently, the discharge-coefficient validation confirms the overall hydraulic performance of the PKW, while the downstream flow-field results require additional consideration of the outlet boundary treatment and the missing experimental bed-slope information.

5.2 Three-Dimensional Air-Water Interface Surface Distribution

The three-dimensional time-averaged water-phase distribution was examined using the mean volume-fraction field, $\alpha_{\text{water,mean}}$. These visualizations provide an overall representation of the air-water interface behaviour throughout the computational domain and allow the influence of the PKW geometry and downstream bed condition to be compared.

5.2.1 Flat-Bed Configurations

The three-dimensional mean air-water interface distributions for the flat-bed LOPKW and NLOPKW configurations are presented in Figures 5.4 and 5.5, respectively. In both cases, considerable air-water interface deformation is observed immediately downstream of the PKW due to the complex overflowing jets. As the flow proceeds farther downstream, the interface gradually becomes smoother and more uniform.

Although the overall downstream behaviour is similar, differences in the near interface surface pattern can be observed between the linear and nonlinear geometries. These differences are associated with the different organization and interaction of the overflowing jets produced by the two PKW configurations.

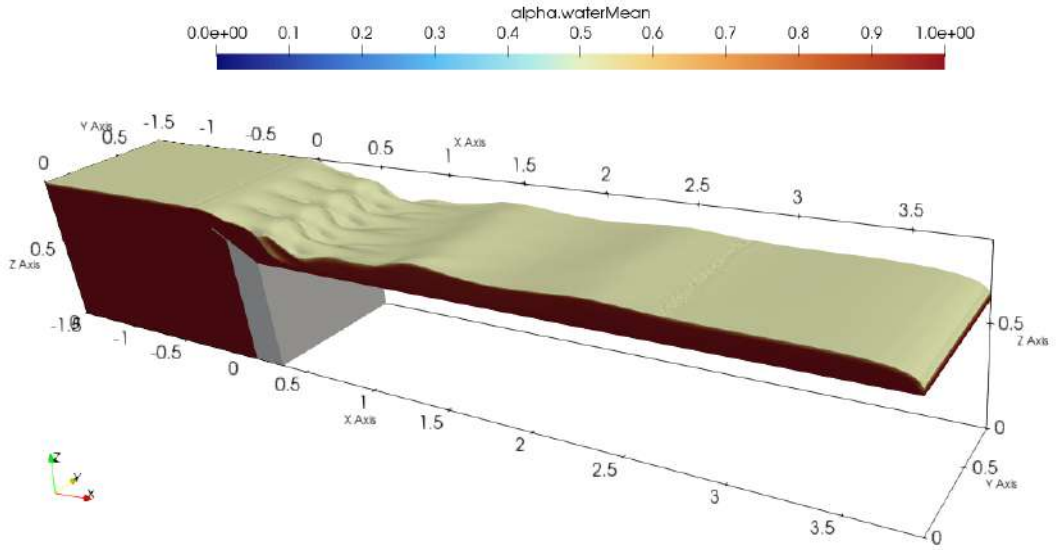


Figure 5.4: Three-dimensional distribution of the time-averaged water volume fraction for the flat-bed linear PKW configuration.

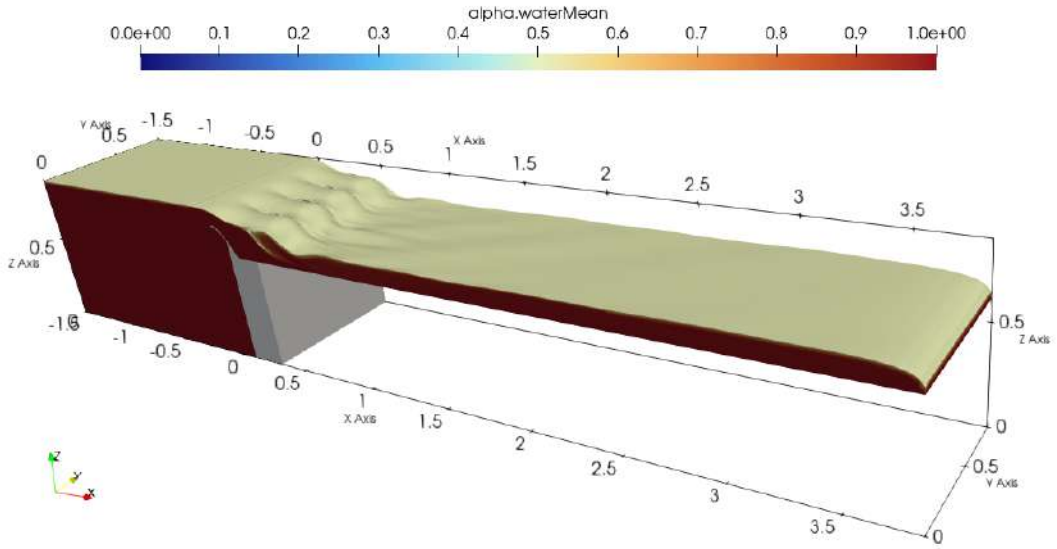


Figure 5.5: Three-dimensional distribution of the time-averaged water volume fraction for the flat-bed nonlinear PKW configuration.

5.2.2 Equilibrium Scour-Bed Configurations

The corresponding three-dimensional distributions for the equilibrium scour-bed LOPKW and NLOPKW configurations are shown in Figures 5.6 and 5.7. Compared with the flat-bed cases, the influence of the equilibrium scour geometry is clearly visible in the downstream region. The deeper scour cavity provides a larger flow region beneath the overflowing jet and modifies the three-dimensional development of the air-water interface surface immediately downstream of the PKW.

For both scour-bed configurations, the strongest air-water interface surface deformation occurs within the near-field region downstream of the structure, while the surface becomes progressively smoother farther downstream. The comparison also demonstrates that the downstream bed morphology plays an important role in controlling the three-dimensional flow development in addition to the PKW geometry itself.

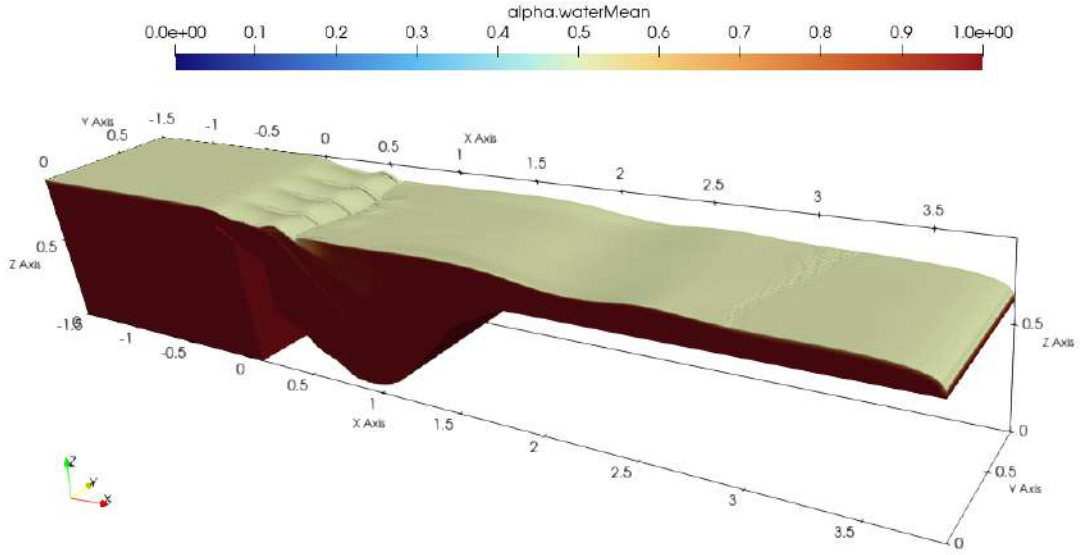


Figure 5.6: Three-dimensional distribution of the time-averaged water volume fraction for the equilibrium scour-bed linear PKW configuration.

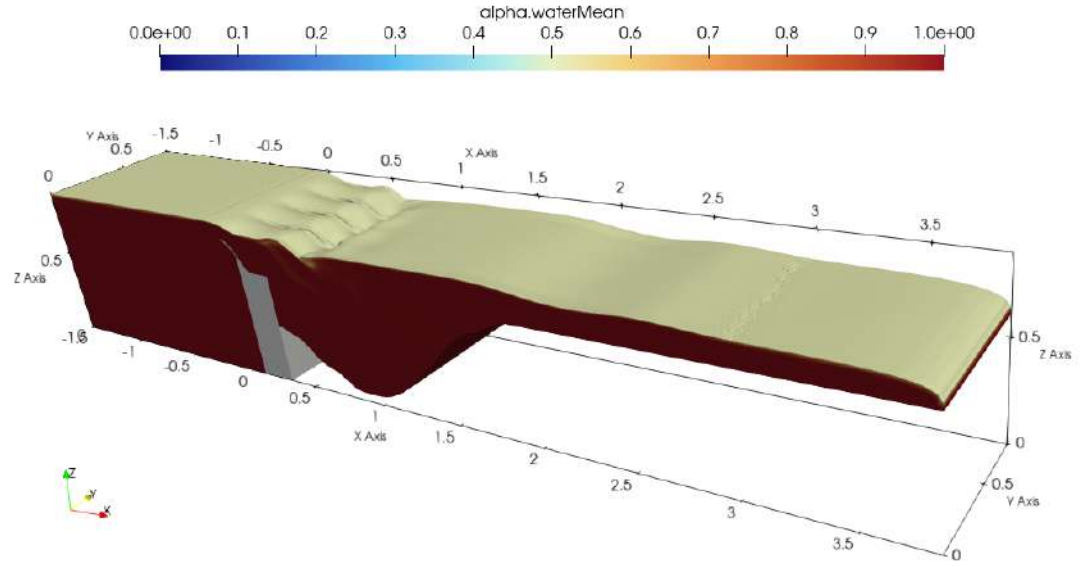


Figure 5.7: Three-dimensional distribution of the time-averaged water volume fraction for the equilibrium scour-bed nonlinear PKW configuration.

5.3 Mean Velocity Field

The time-averaged velocity-magnitude field was used to investigate and examine the high velocity region on the surface of PKW and downstream region of PKW. The mean field eliminates fluctuations in the flow and gives better demonstrations of hydraulic behavior of flow during statistically quasi-steady period. The identical velocity scale was used for direct comparison between all numerical cases.

5.3.1 Flat-Bed Configurations

The mean velocity-magnitude fields for the flat-bed linear and nonlinear PKWs are shown in Figures 5.8 and 5.9, respectively. In both configurations, the flow accelerates while passing over the PKW and forms a high-velocity region immediately downstream of the structure.

If we compare both the LOPKW and NLOPKW mean velocity fields, we can find that in LOPKW, the velocity was high in the region of the PKW surface and in the downstream region, which is close to the PKW. In the linear case, velocity persists for a very short term throughout the downstream and vanishes as we proceed along the downstream bed. But in the case of NLOPKW, velocity in the downstream region extends far to the downstream region. By comparing both the cases, we found that the velocity gradient in the Nonlinear case was more as compared to the Linear case as illustrated in Figures 5.8 and 5.9.

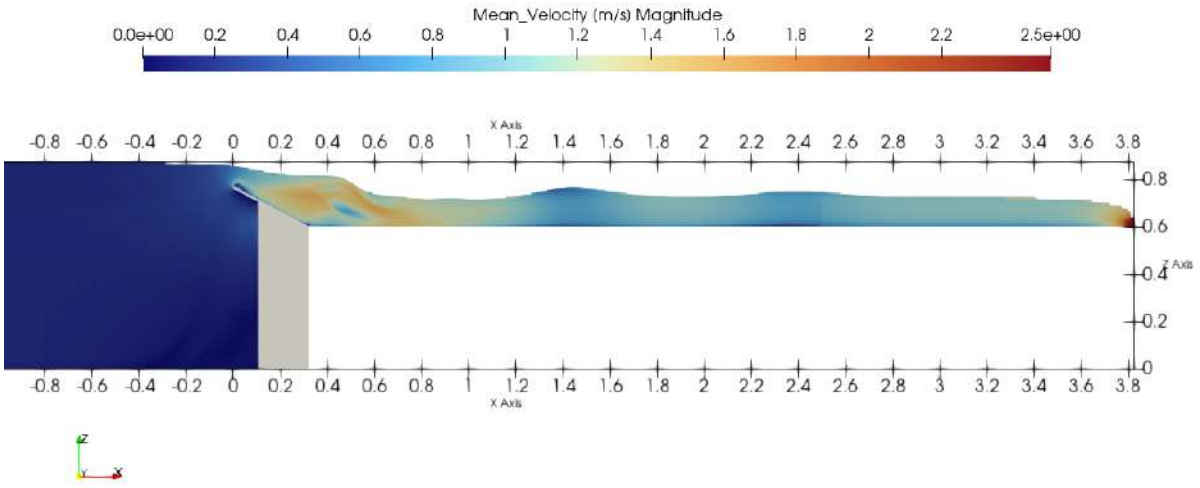


Figure 5.8: Time-averaged velocity-magnitude field for the flat-bed linear PKW configuration.

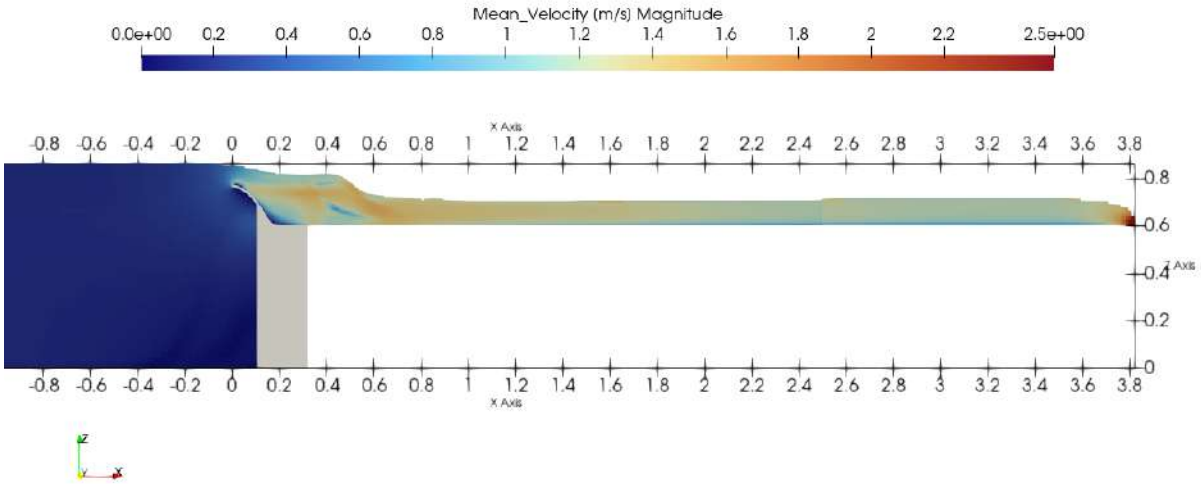


Figure 5.9: Time-averaged velocity-magnitude field for the flat-bed nonlinear PKW configuration.

5.3.2 Equilibrium Scour-Bed Configurations

The mean velocity field for the equilibrium scour-bed linear configuration And Nonlinear configuration is illustrated in Figure 5.10 and 5.11.

In both cases, the peak mean velocity occurs just near the PKW at downstream bed where the overflow jets enter the scour region. We can notice that the main high velocity flow occurs above the scour region whereas the low velocity regions occur in the deeper scour cavity causing momentum to transfer along vertical direction. In comparison with the LOPKW, NLOPKW depicts a slightly extended high velocity region along the upper part of scour cavity. The interaction between high velocity flow layer and low velocity flow layer may develop strong turbulence motion and may affect the bed shear distribution across channel bed.

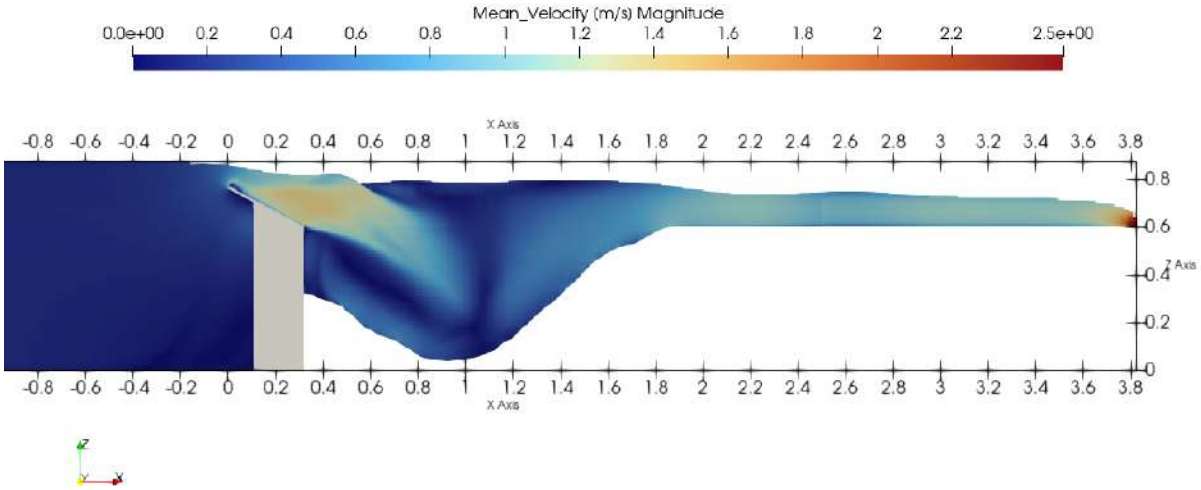


Figure 5.10: Time-averaged velocity-magnitude field for the equilibrium scour-bed linear PKW configuration.

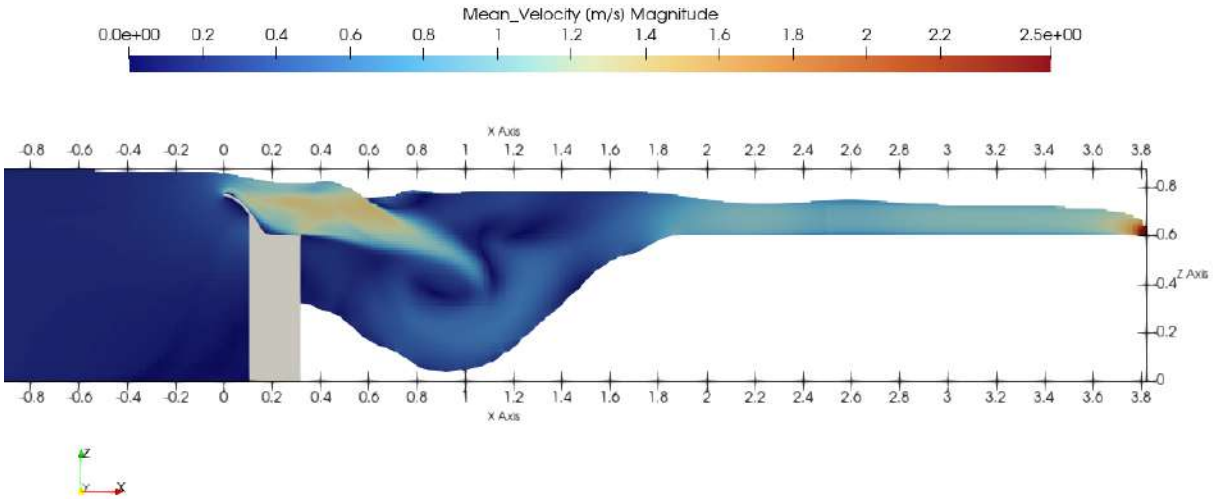


Figure 5.11: Time-averaged velocity-magnitude field for the equilibrium scour-bed nonlinear PKW configuration.

5.4 Streamline Patterns and Recirculation Zones

Streamlines provide additional information about flow direction, separation and recirculation that cannot be obtained from velocity magnitude alone. They are therefore used to identify vortical structures, the direction of the main jet and the development of downstream recirculation zones.

5.4.1 Flat-Bed Configurations

The streamline patterns for the flat-bed linear and nonlinear PKWs are presented in Figures 5.12 and 5.13. These streamlines provide much clearer visualization of how flow propagates over the PKW and highlights the vortices and recirculation region if it exists.

The streamline patterns for the flat-bed linear and nonlinear PKWs are presented in Figures 5.12 and 5.13. Both figures illustrate the formation of vortices just underneath the PKW, which was found to be very interesting, as these vortices are a source of high energy dissipation and enhance turbulent energy mixing. These vortical structures, with time, can become sources of local bed shear and, in the end, could contribute to the scouring process.

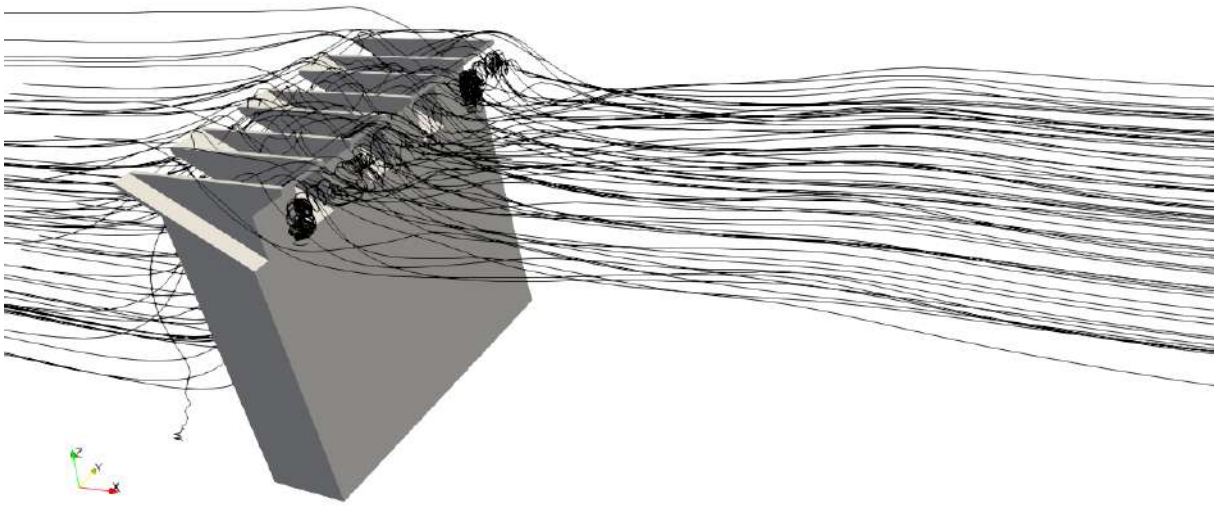


Figure 5.12: Three-dimensional streamline pattern for the flat-bed linear PKW configuration.

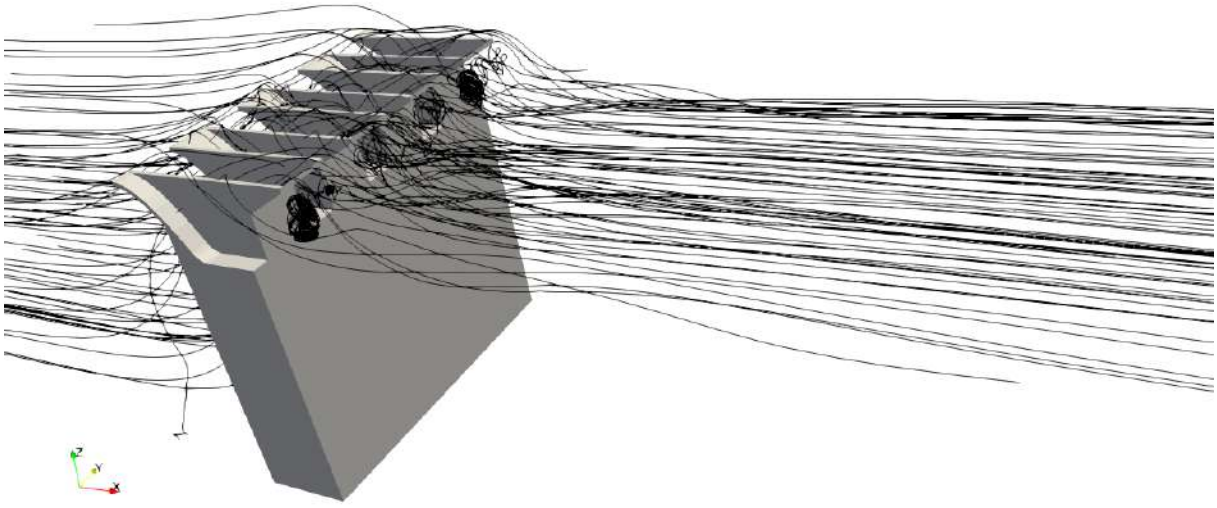


Figure 5.13: Three-dimensional streamline pattern for the flat-bed nonlinear PKW configuration.

5.4.2 Equilibrium Scour-Bed Configurations

The streamline patterns for the equilibrium scour-bed linear and nonlinear PKW configurations are presented in Figures 5.14 and 5.15.

In both LOPKW and NLOPKW, a primary and prominent recirculation region is produced within the deeper scour cavity region at downstream region just near the PKW. If we carefully visualize the streamlines, we can notice that main flow passes over the upper region of scour hole while the flow within scour region forms recirculating vortex. The differences in velocities between main downstream flow and recirculating flow created a shear effect between layers and produces a shear force along the upper boundary of vortex which can contribute to high turbulent kinetic energy and momentum exchange. Farther downstream, flow becomes gradually aligned with the channel bed. NLOPKW illustrated relatively bigger recirculation region as compared to LOPKW, but the strength of vortex cannot be claimed directly by visualizing the streamlines instead it should be supported by velocity contour, Turbulent kinetic energy (TKE), vorticity etc.

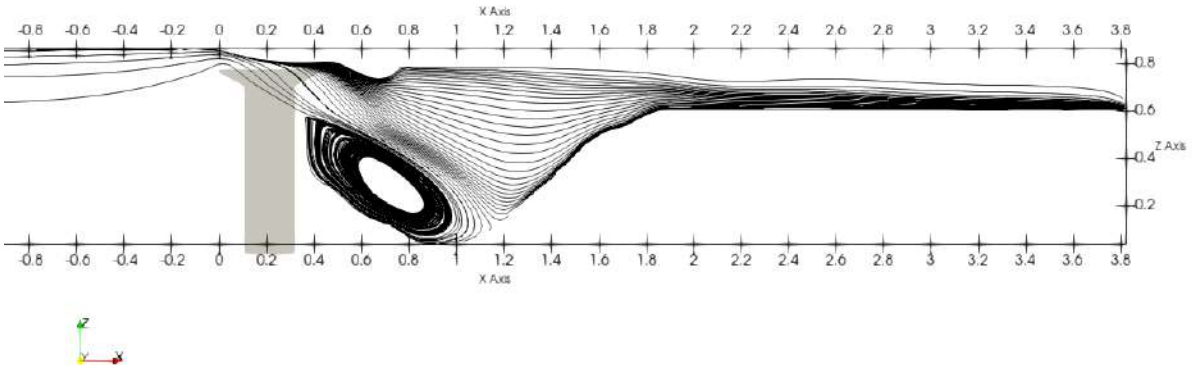


Figure 5.14: Streamline pattern within the equilibrium scour cavity for the linear PKW configuration.

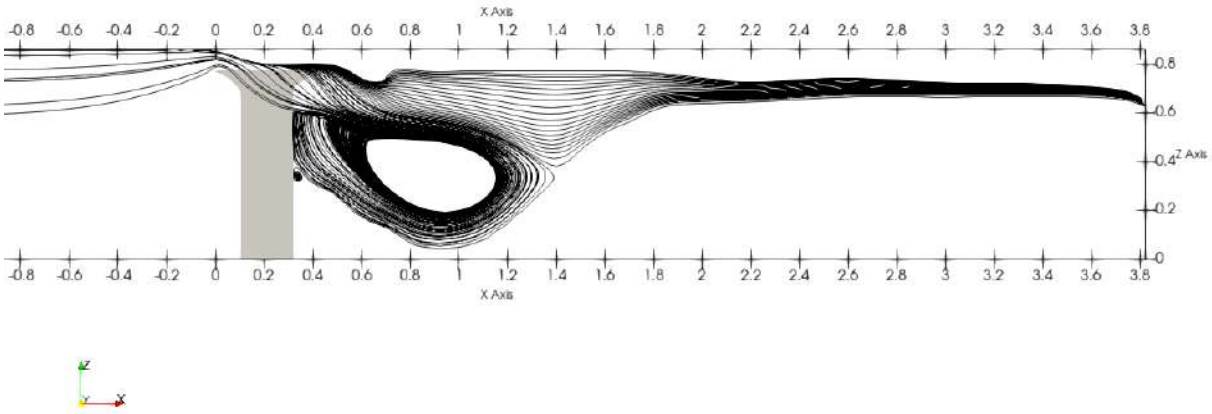


Figure 5.15: Streamline pattern for the equilibrium scour-bed nonlinear PKW configuration.

5.5 Mean Turbulent Kinetic Energy

The mean turbulent kinetic energy (TKE) field was analysed to identify regions of enhanced turbulent activity. Elevated TKE is expected where strong velocity gradients, jet spreading, flow separation and recirculation occur. The turbulence results are therefore interpreted together with the mean velocity and streamline distributions.

5.5.1 Flat-Bed Configurations

The mean turbulent kinetic energy distribution for the flat-bed linear and nonlinear cases is illustrated in Figures 5.16 and 5.17. We can notice that, for both cases, the peak turbulent kinetic energy occurs just near the PKW outlet keys in the downstream region, while farther downstream, the turbulent kinetic energy is very low.

For the LOPKW, we can notice that the peak mean turbulent kinetic energy is concentrated in the regions just near the outlet of the PKW, indicating energetic fluctuations and high turbulent mixing, and decreases as we go farther downstream. Whereas in the NLOPKW, the peak mean turbulent kinetic energy also occurs just near the outlet keys, similar to the LOPKW, but with the difference that it is more spread and diffused and extends farther downstream as compared to the LOPKW.

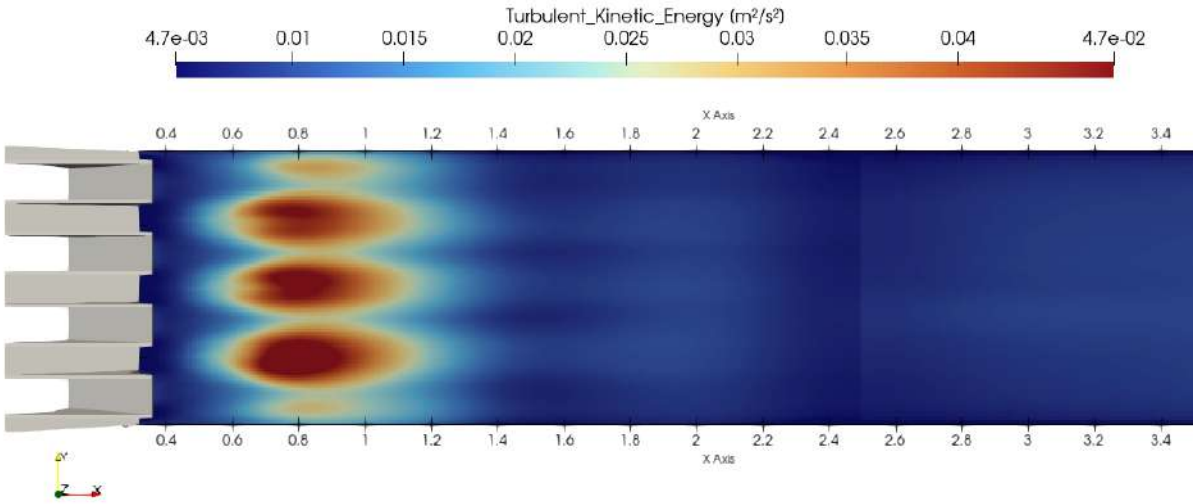


Figure 5.16: Mean turbulent kinetic energy distribution for the flat-bed linear PKW configuration.

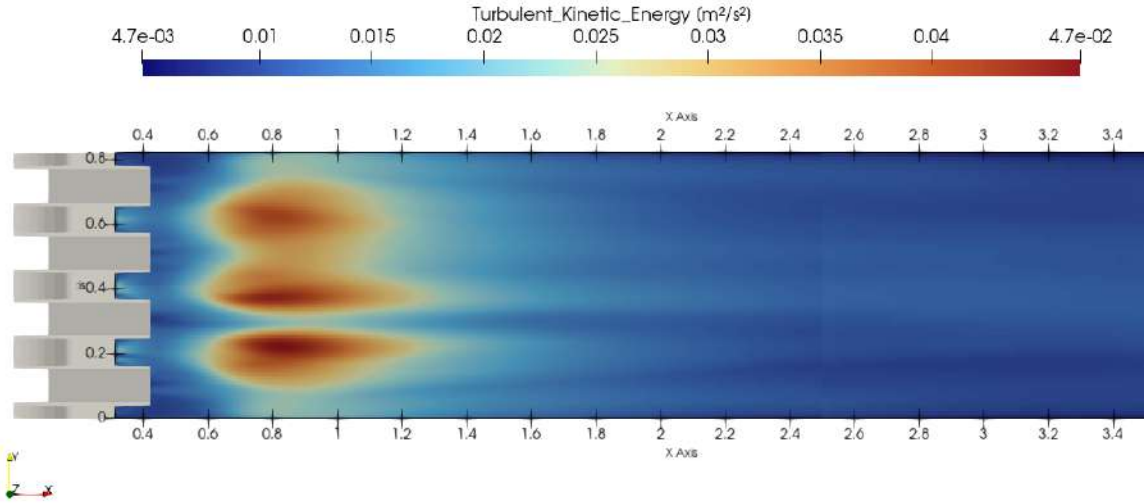


Figure 5.17: Mean turbulent kinetic energy distribution for the flat-bed nonlinear PKW configuration.

5.5.2 Equilibrium Scour-Bed Configurations

The mean turbulent kinetic energy distributions for the equilibrium scour-bed linear and nonlinear PKW configurations are presented in Figures 5.18 and 5.19.

For both configurations, relatively high turbulent kinetic energy was observed at the downstream region along the scour cavity, just near the PKW. This region corresponds to the recirculation region as observed in the previous streamlines section where interaction between main downstream flow and recirculating flow creates shear effect and turbulent mixing. Compared to LOPKW, NLOPKW demonstrates turbulent kinetic energy farther downstream whereas in LOPKW, The peak Turbulent kinetic energy is observed closer to PKW.

In both configurations, turbulence kinetic energy gradually vanishes as flow approaches farther downstream region resulting in dissipation of turbulence and reattachment of flow along channel bed.

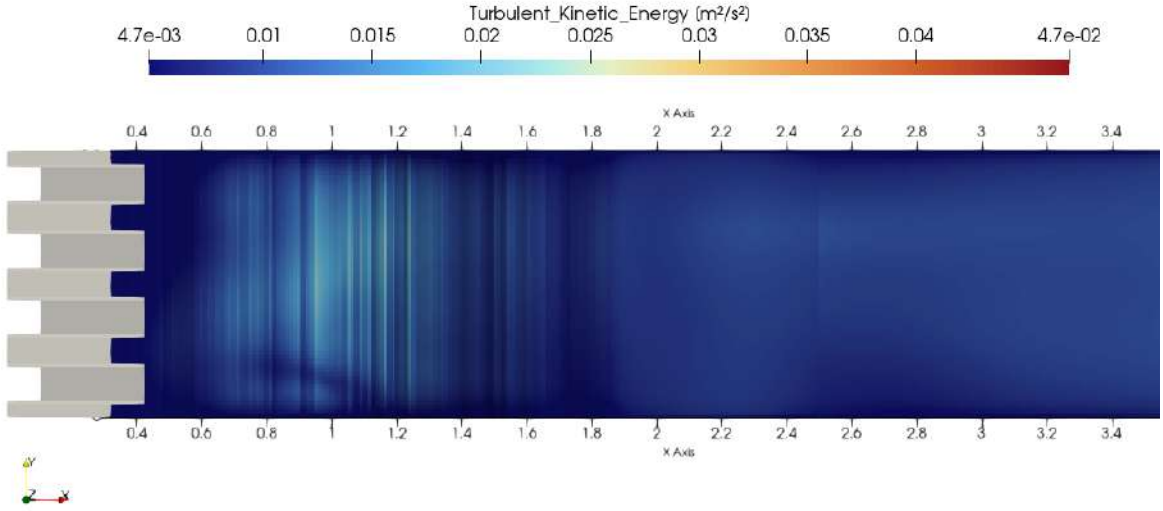


Figure 5.18: Mean turbulent kinetic energy distribution for the equilibrium scour-bed linear PKW configuration.

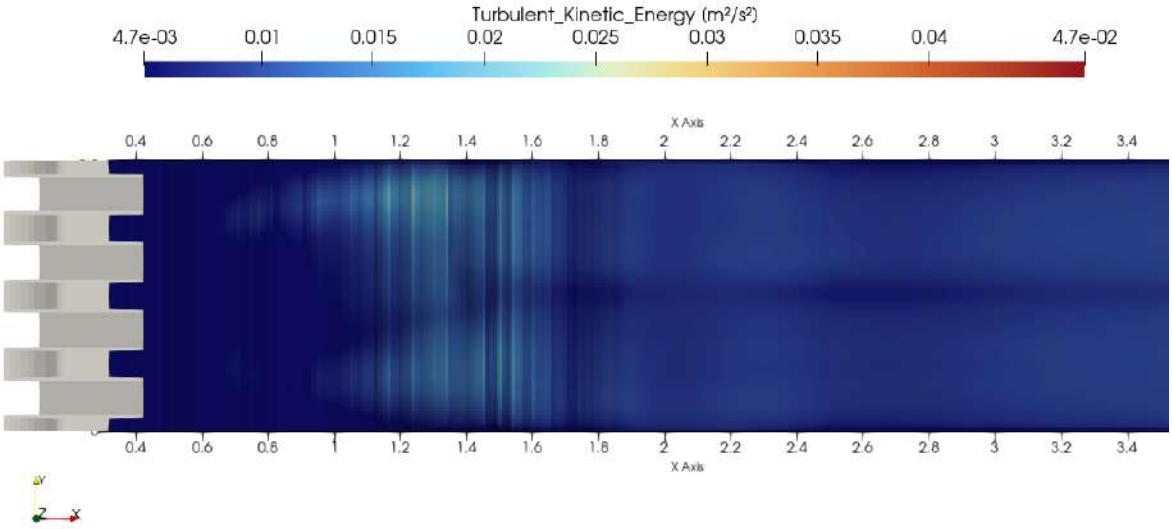


Figure 5.19: Mean turbulent kinetic energy distribution for the equilibrium scour-bed nonlinear PKW configuration.

5.6 Bed Shear Stress Distribution

Bed shear stress was examined because it represents the tangential hydrodynamic loading acting on the downstream bed and is directly relevant to the potential initiation of sediment motion. Both spatial contour plots and quantitative longitudinal profiles were used.

The contour plots provide an overview of the spatial distribution of the time-averaged bed shear stress, whereas the longitudinal plots provide a direct comparison between the linear and nonlinear geometries at selected sampling locations.

5.6.1 Flat-Bed Configurations

The mean bed shear-stress distributions for the flat-bed LOPKW and NLOPKW configurations as illustrated in Figures 5.20 and 5.21 demonstrates that highest bed shear stresses dominate in the immediate downstream region just near the outlet key where the jets strike abruptly with the downstream bed.

In the LOPKW case, the peak bed shear stresses exist just near the PKW and decreases rapidly along the downstream bed whereas in NLOPKW, the mean bed shear stresses extend father more in downstream region which means that NLOPKW exhibits intense interaction between downstream flow bed and channel bed over a long downstream distance.

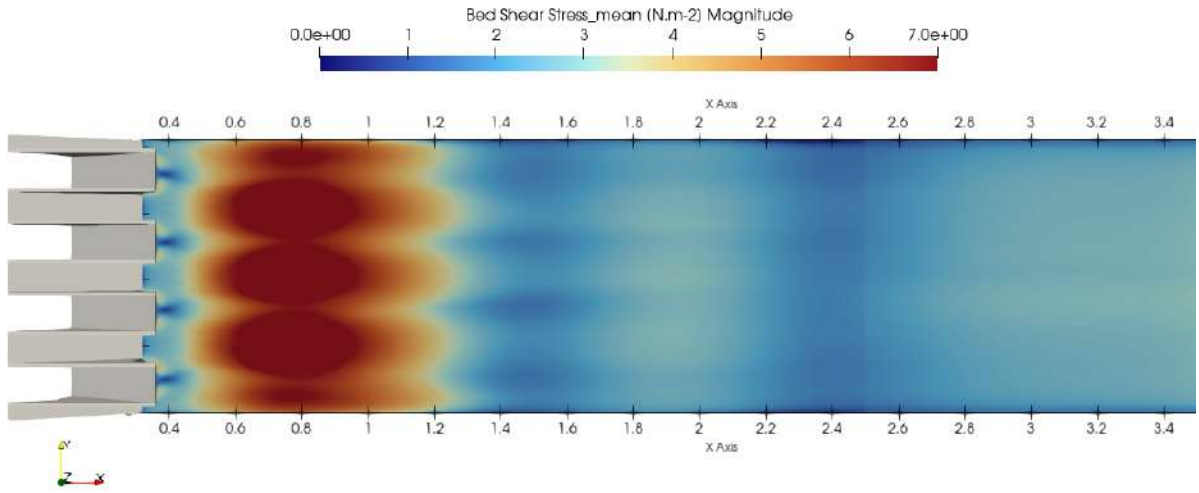


Figure 5.20: Top view of the mean bed shear-stress magnitude for the flat-bed linear PKW configuration.

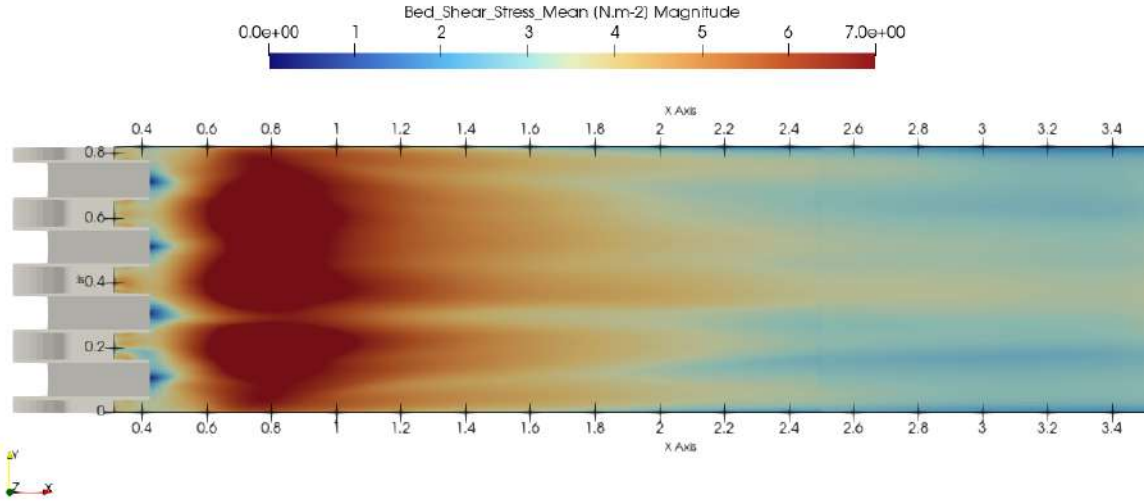


Figure 5.21: Top view of the mean bed shear-stress magnitude for the flat-bed nonlinear PKW configuration.

5.6.2 Quantitative Comparison of Flat-Bed PKWs

A quantitative comparison of the longitudinal mean bed shear stresses was analyzed along with the inlet-key, outlet-key and side-wall locations. The critical was calculated based on the experimental measurements and comes out to be $\tau_c = 4.90\ \mathrm{N/m^2}$ which is represented by dotted horizontal line as indicated in all plots.

For the inlet-key profile as illustrated in Figure 5.22, both LOPKW and NLOPKW exhibits a peak mean bed shear stress of around approximately 7 $\mathrm{N/m^2}$ within the downstream bed region of around 0.45–0.50 m. These values surpass the critical shear stress needed to initiate the movement of sediment particles of bed resulting in initiation of scouring phenomenon. Beyond the peak region, the mean bed shear stress for the LOPKW decreases relatively fast and drops the critical value over a short downstream period as compared to NLOPKW which shows a gradual decay in mean bed shear stresses and elevated mean bed shear stresses persist farther downstream for NLOPKW.

The outlet-key profile comparison is illustrated in Figure 5.28. Outlet-key produces a peak bed shear stress among all three profiles. The LOPKW approaches a maximum mean bed shear stress of around 8.8 $\mathrm{N/m^2}$ whereas for NLOPKW, it comes out to be 8.2 $\mathrm{N/m^2}$. After crossing the peak shear stresses values, NLOPKW demonstrates a gradual decay in mean bed shear stresses and persists farther downstream as compared to LOPKW which depicts the sudden decay in mean bed shear stress value and drops below the critical shear stress value over a short downstream distance.

The sidewall profile comparison is illustrated in Figure 5.24. The sidewall profile shows trend like inlet and outlet keys with peak mean bed shear stress of around 7 $\mathrm{N/m^2}$ just near the PKW downstream. NLOPKW demonstrates a gradual decay in mean bed shear stresses and

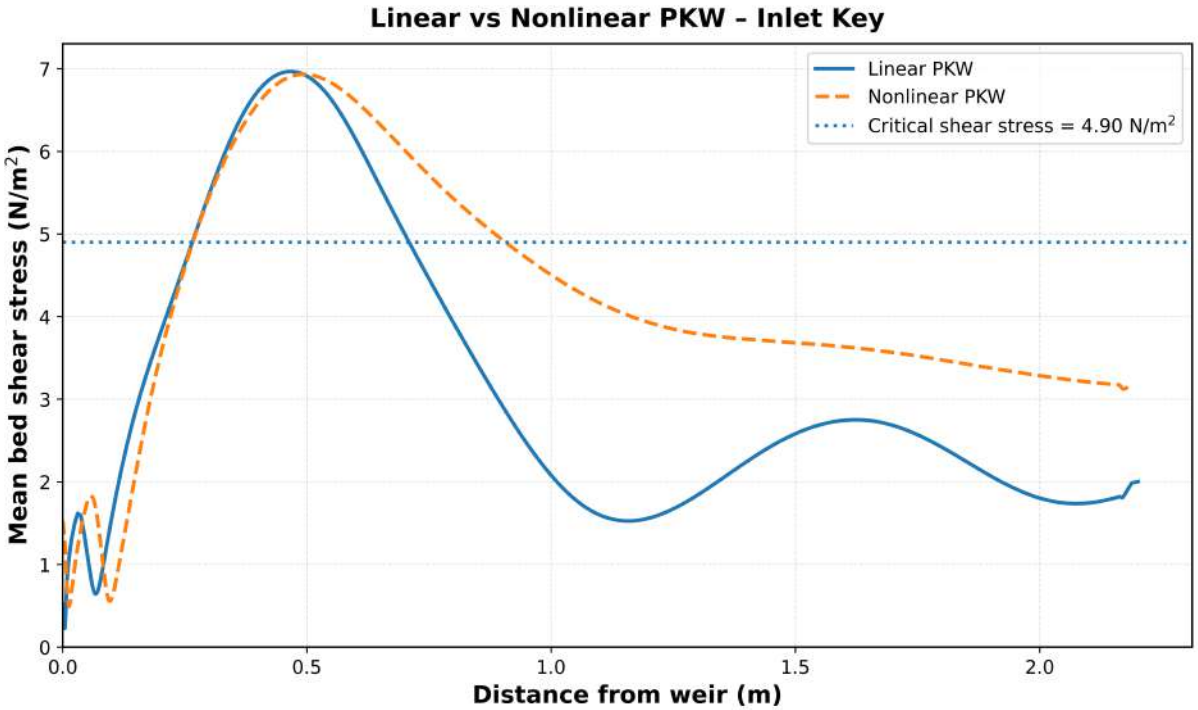


Figure 5.22: Comparison of mean bed shear stress along the inlet-key sampling location for the flat-bed linear and nonlinear PKWs.

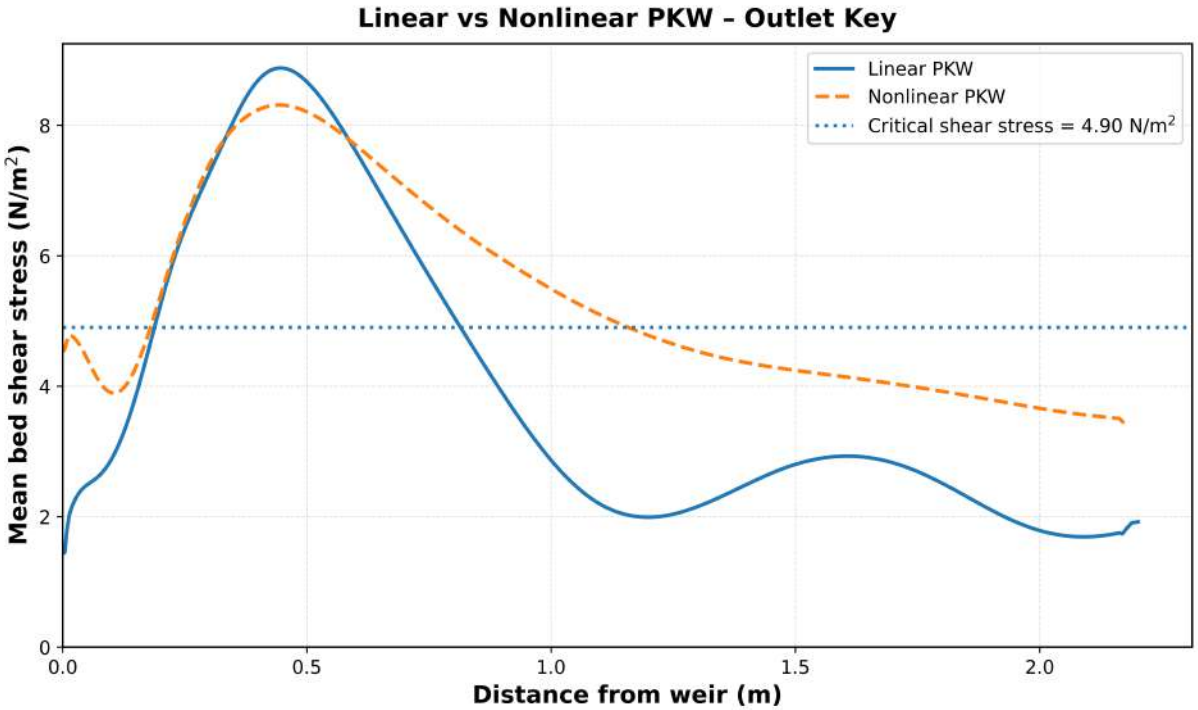


Figure 5.23: Comparison of mean bed shear stress along the outlet-key sampling location for the flat-bed linear and nonlinear PKWs.

persists farther downstream as compared to LOPKW which depicts the sudden decay in mean bed shear stress value and drops below the critical shear stress value over a short downstream distance.

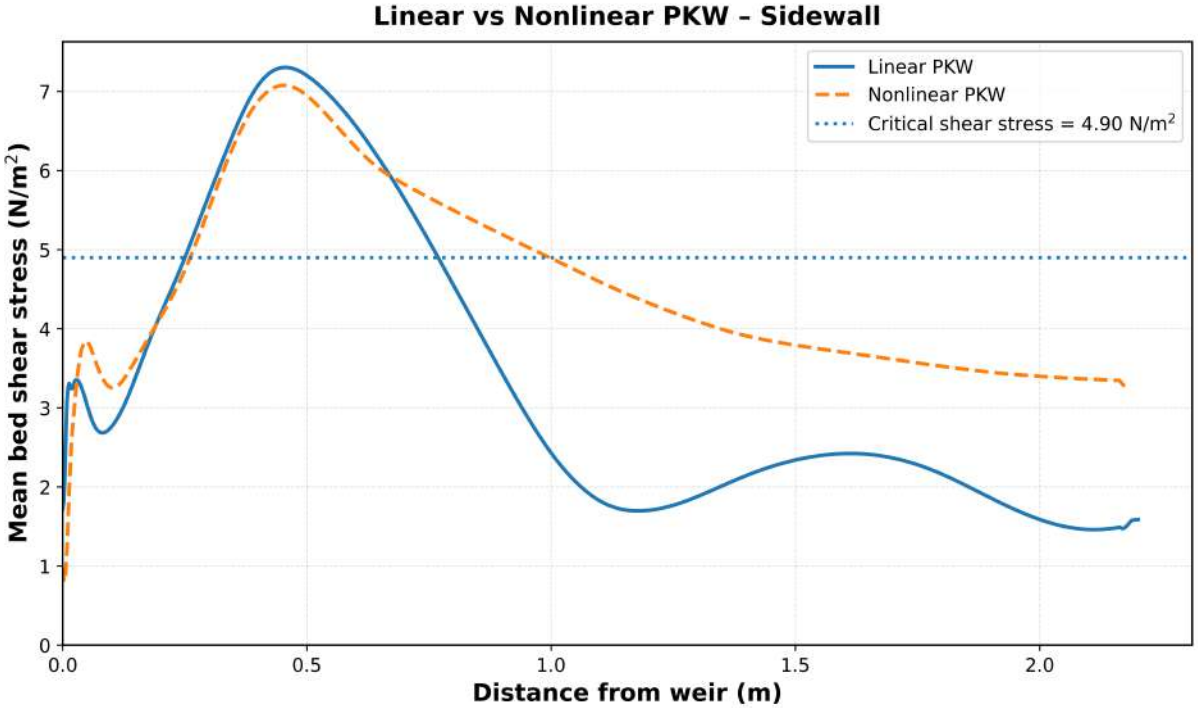


Figure 5.24: Comparison of mean bed shear stress along the side-wall sampling location for the flat-bed linear and nonlinear PKWs.

5.6.3 Equilibrium Scour-Bed Configurations

The time-averaged bed shear-stress magnitude distributions for the equilibrium scour-bed linear and nonlinear PKW configurations are presented in Figures 5.25 and 5.26

For both configurations, relatively low mean bed shear stresses are observed at the downstream region along the scour cavity, just near the PKW. This region corresponds to the recirculation region as observed in the previous streamlines section where interaction between main downstream flow and recirculating flow creates shear effect and turbulent mixing. Farther downstream, the mean bed shear stresses become high as the flow approaches farther downstream, and flow becomes more aligned along the channel bed.

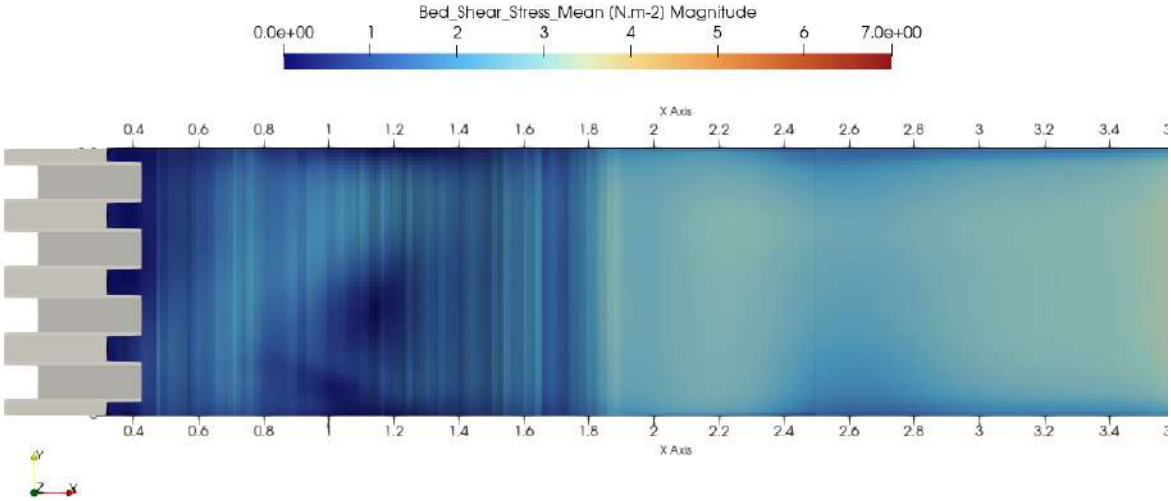


Figure 5.25: Top view of the mean bed shear-stress magnitude for the equilibrium scour-bed linear PKW configuration.

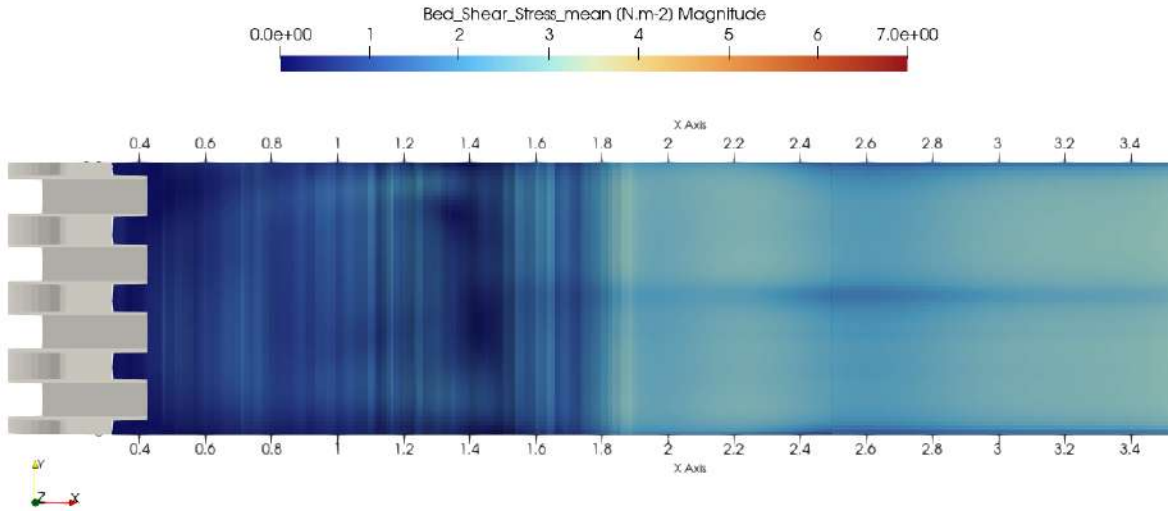


Figure 5.26: Top view of the mean bed shear-stress magnitude for the equilibrium scour-bed nonlinear PKW configuration.

5.6.4 Linear–Nonlinear Comparison for the Equilibrium Scour Bed

Figure 5.27, Figure 5.28 and Figure 5.29 compare the mean shear-stress distributions for the equilibrium scour-bed LOPKW and NLOPKW configurations at the inlet-key, outlet-key, and side-wall profile.

LOPKW generally exhibits high mean stress than the NLOPKW at the inlet key profile and outlet key profile whereas NLOPKW generally exhibits higher mean stresses at side wall profile over along portion of downstream region. For both configurations, low over all bed shear stresses were reported along the scour cavity whereas relatively high bed shear stresses were reported at farther downstream region with a downstream distance e of around x = 1.8–1.9 m corresponding to downstream flow reattachment along the bed channel. In comparison with flat bed case, equilibrium scour bed case reported overall very low mean bed shear stresses along the complete downstream region.

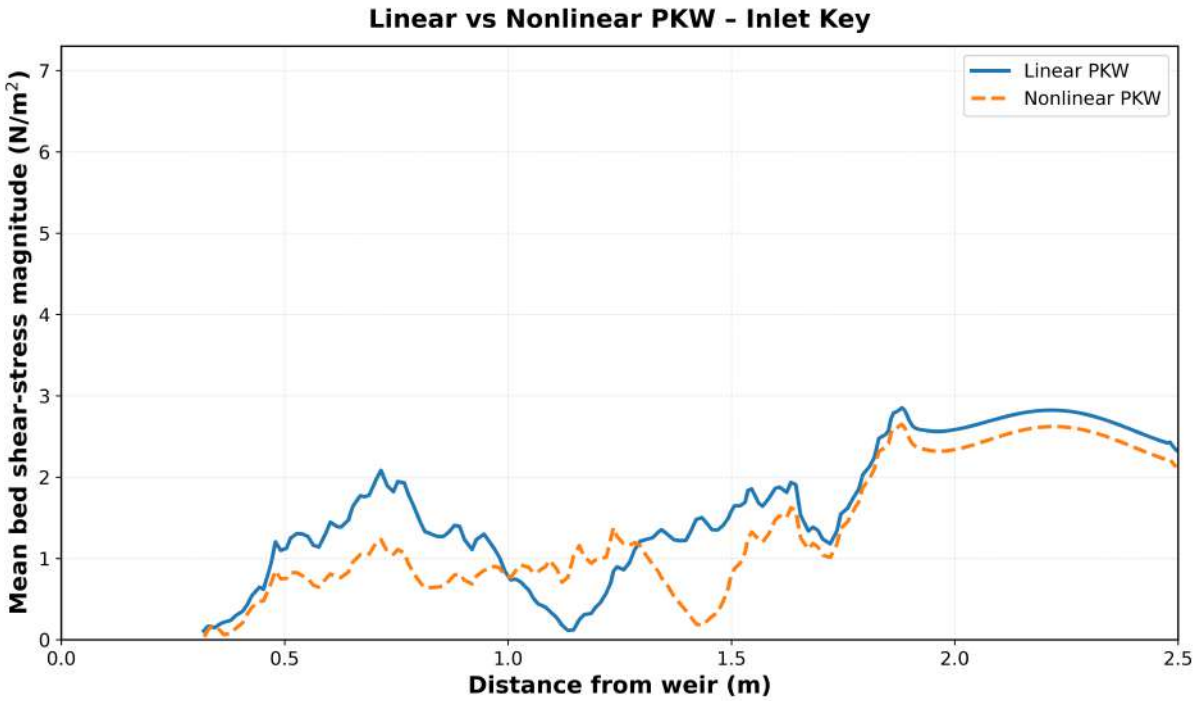


Figure 5.27: Comparison of mean bed shear stress along the inlet-key location for the equilibrium scour-bed configurations.

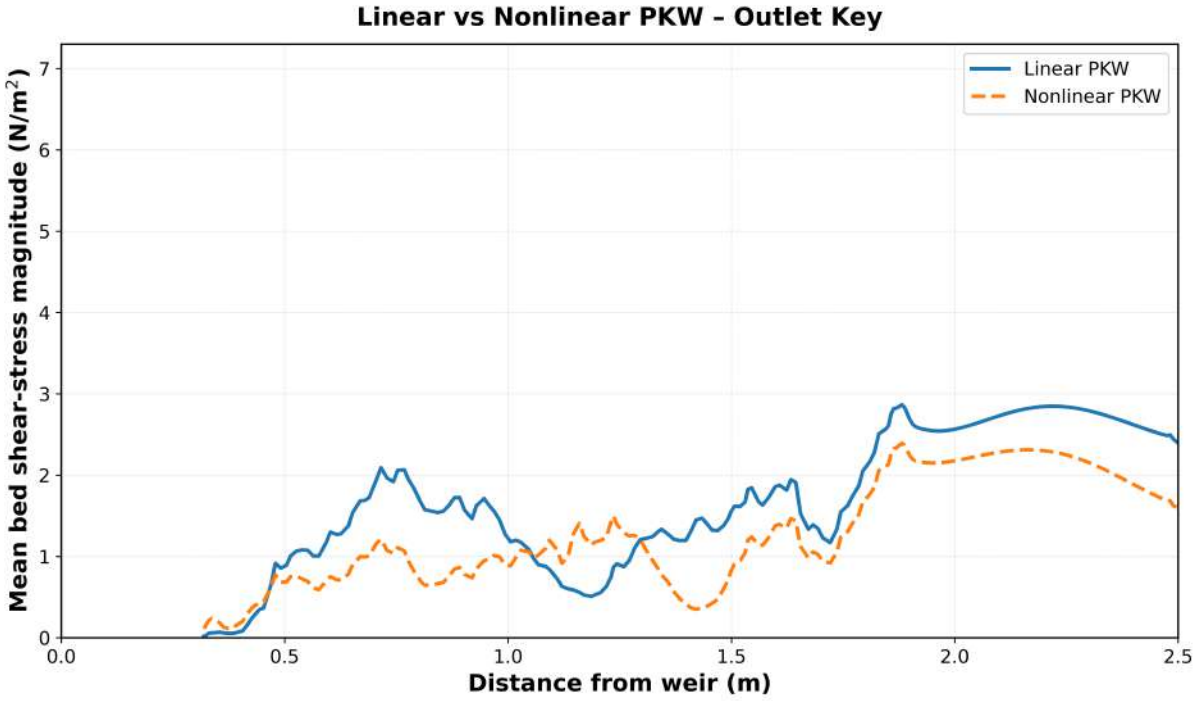


Figure 5.28: Comparison of mean bed shear stress along the outlet-key location for the equilibrium scour-bed configurations.

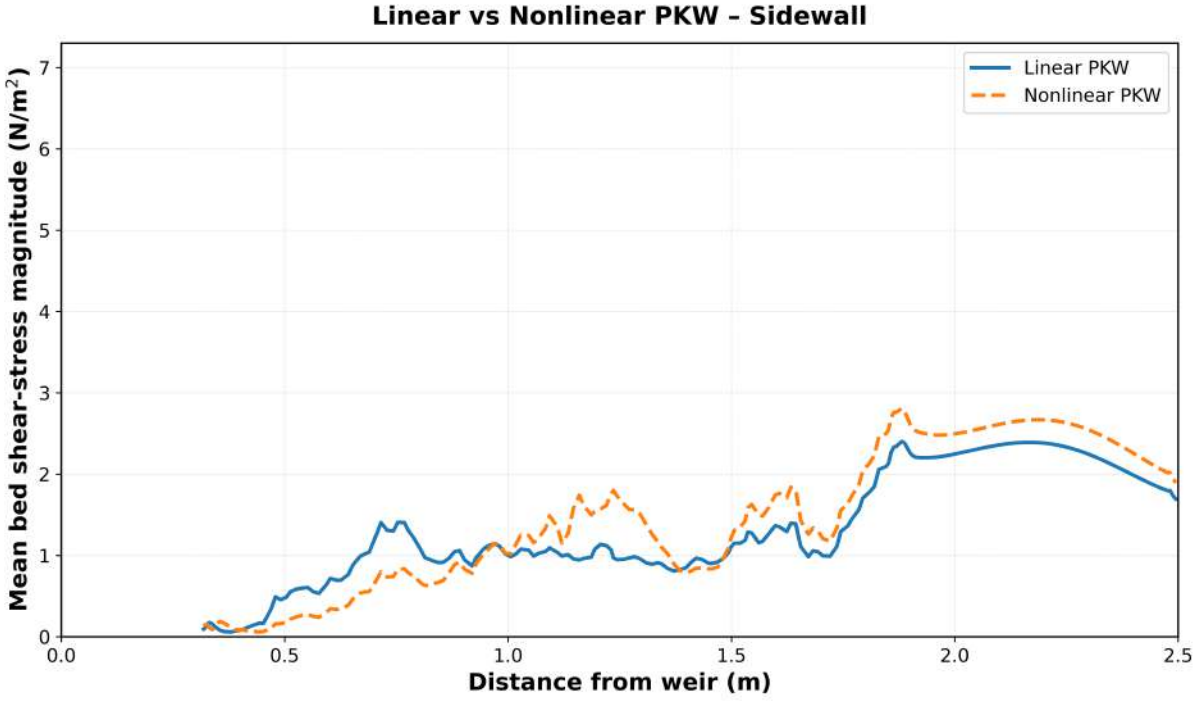


Figure 5.29: Comparison of mean bed shear stress along the side-wall location for the equilibrium scour-bed configurations.

5.7 Discussion of Main Findings

The numerical results demonstrate that the downstream hydraulic response of a Piano Key Weir is controlled by the combined effects of PKW geometry and downstream bed configuration. Changes in the outlet-key geometry modify the organization of the overflowing jets, which subsequently influences the mean velocity distribution, turbulent kinetic energy, recirculation structure, free-surface behaviour, and near-bed shear stress.

For the flat-bed configurations, the largest bed shear stresses were concentrated in the near-field region immediately downstream of the PKW. The longitudinal shear-stress profiles showed that the LOPKW generally produced slightly larger local peak values, whereas the NLOPKW exhibited a slower downstream decay and maintained elevated shear stress over a greater distance. Therefore, the hydraulic effect of the PKW geometry cannot be assessed from the maximum shear stress alone; the downstream persistence of the applied stress is also important when considering potential sediment mobilisation.

The equilibrium scour-bed configurations exhibited a substantially different hydraulic response. The greater local flow depth within the scour cavity allowed the development of a pronounced primary recirculation zone beneath the principal overflowing jet. As a result, the main high-velocity flow was located predominantly above the recirculating region rather than acting directly along the deepest part of the scour surface. This modification of the flow structure was also reflected in the turbulent kinetic energy and bed shear-stress distributions.

Comparison of the equilibrium scour-bed LOPKW and NLOPKW configurations showed that the LOPKW generally produced higher mean bed shear stresses along the inlet-key and outlet-key profiles, whereas the NLOPKW showed relatively higher values along part of the side-wall profile. For both configurations, the mean bed shear stress was comparatively low along the substantial of the scour cavity. A local increase was observed farther downstream at approximately $x = 1.8$–1.9 m, which corresponds to the region where the downstream flow approaches reattachment with the raised bed. Most importantly, the overall bed shear-stress levels for the equilibrium scour-bed configurations were substantially lower than those obtained for the corresponding flat-bed cases.

This reduction supports the principal physical interpretation of the study. The initially high hydraulic loading contributes to conditions favourable for sediment mobilisation, while the resulting scour geometry increases the local flow depth and modifies the jet and recirculation structure. The equilibrium bed therefore experiences considerably lower overall hydraulic loading than the initial flat bed. The calculated critical shear stress of $\tau_c = 4.90\ \mathrm{N/m^2}$ was used as an engineering reference for this assessment and not as an exact deterministic boundary for sediment motion.

The numerical model also reproduced the upstream hydraulic performance with good agreement. For the flat-bed cases, the relative differences between experimental and numerical discharge coefficients were 7.35% for the LOPKW and 4.20% for the NLOPKW. Even closer

agreement was obtained for the equilibrium scour-bed configurations, for which the corresponding differences were approximately 0.77% and 0.81%, respectively.

Although the upstream hydraulic behaviour was reproduced well, larger differences were observed for the downstream water level. The numerical downstream water elevations were generally higher than the corresponding experimental values. One limitation is that the longitudinal slope of the experimental downstream channel bed was not available and therefore could not be reproduced in the numerical domain. Furthermore, the downstream water level was not prescribed directly; instead, the outlet boundary was selected to allow the flow to evolve naturally. Consequently, the missing experimental bed slope and the outlet treatment may both have contributed to the difference in downstream water depth.

A further numerical feature requiring attention was the local acceleration of the flow and reduction in water depth close to the outlet. This behaviour was visible across the numerical results and suggests that the downstream solution may remain sensitive to the outlet boundary treatment and/or the available downstream development length. The interpretation of the flow very close to the outlet should therefore be made with caution.

Chapter 6

Conclusion

This thesis numerically investigated hydraulic bahviour of flow over Linear Outlet Piano Key Weirs (LOPKWs) and Non-Linear Outlet Piano Key Weirs (NLOPKWs) by changing the PKW geometry and using flat bed and equilibrium scoured bed configuration by using three-dimensional CFD simulations in OpenFOAM. Four numerical configurations were considered by combining the two PKW geometries with an initial flat downstream bed and experimentally measured equilibrium scour-bed profiles. The air–water interface was modelled using the VOF method with `interFoam`, while turbulence was represented using the RANS k–ω SST model. The equilibrium scour geometries were imposed as fixed experimentally measured bed profiles; therefore, sediment transport and the temporal development of scour were not simulated directly.

Statistically quasi-steady conditions were established before the final time-averaged analysis. The monitored inlet and outlet discharges, total water volume and upstream water level showed no significant long-term drift during the selected averaging interval. The model also reproduced the experimental upstream hydraulic behaviour satisfactorily. The differences in discharge coefficient for the flat-bed LOPKW and NLOPKW were 7.35% and 4.20%, respectively, while the equilibrium scour-bed cases showed considerably closer agreement, with differences of approximately 0.77% and 0.81%.

The numerical results demonstrated clear differences between the flat-bed and equilibrium scour-bed hydraulic conditions. For the flat-bed cases, high-velocity jets and vortical structures were generated immediately downstream of the PKW. These regions were associated with high turbulent activity and comparatively large bed shear stresses. The linear PKW generally produced slightly larger local peak shear stresses, while the nonlinear PKW tended to maintain elevated stress farther downstream.

In the equilibrium scour-bed cases, the deeper downstream geometry produced a pronounced primary recirculation zone within the scour cavity. The main flow passed predominantly above this recirculating region, resulting in substantially lower overall bed shear stresses within the scour cavity than those observed for the initial flat-bed configurations. A local increase in shear stress occurred farther downstream near $x = 1.8$–1.9 m, where the flow approached reattachment with the raised downstream bed. These results support the main hypothesis of the study that

development of an equilibrium scour geometry modifies the flow field in a manner that reduces the overall hydraulic loading acting on the bed.

The comparison between the linear and nonlinear equilibrium scour-bed cases showed that the LOPKW generally produced larger shear stresses along the inlet-key and outlet-key profiles, whereas the NLOPKW produced relatively higher stresses along part of the side-wall profile. The results therefore show that both the PKW geometry and the downstream bed morphology affect the spatial distribution of hydraulic loading.

Some limitations remain in the downstream hydraulic prediction. The numerical downstream water levels were generally higher than the experimental values, while the experimental longitudinal downstream bed slope was not available for inclusion in the computational geometry. Furthermore, a high-velocity region accompanied by a reduction in water depth was observed near the numerical outlet. Since the downstream water level was allowed to evolve naturally rather than being prescribed, these effects may be associated with the missing experimental bed slope, outlet boundary treatment, downstream domain length, or a combination of these factors.

Overall, the study demonstrates that three-dimensional CFD can provide detailed information on velocity, turbulence, vortical structures, recirculation and bed shear stress that complements experimental measurements. The comparison between the initial flat bed and experimentally measured equilibrium scour bed provides useful insight into how modification of the downstream morphology changes the hydraulic conditions around Piano Key Weirs.

Chapter 7

Future Perspectives

The results of the present study identify several areas where further investigation would improve the numerical prediction of the downstream hydraulics of Piano Key Weirs.

A primary requirement is the accurate measurement and representation of the longitudinal slope of the experimental downstream channel bed. This information was unavailable in the present study and may have contributed to the difference between the numerical and experimental downstream water levels. Future experimental investigations should therefore record the complete downstream channel geometry in addition to the local equilibrium scour profile.

The behaviour near the numerical outlet also requires further investigation. A local increase in velocity accompanied by a reduction in water depth was observed close to the downstream boundary. Future work should therefore perform sensitivity studies using increased downstream domain lengths and alternative outlet boundary treatments. The present natural-outflow approach could also be compared with simulations in which the experimentally measured tailwater level is prescribed. Such comparisons would help determine whether the observed outlet behaviour results primarily from the boundary condition, domain length, downstream bed slope, or their combined influence.

Additional experimental measurements would provide a stronger validation of the local CFD results. The current model was successfully assessed using upstream hydraulic conditions and discharge coefficients, but detailed experimental measurements of velocity and turbulence in the downstream region were limited. Techniques such as Particle Image Velocimetry (PIV) or Acoustic Doppler Velocimetry (ADV) could be used to validate the predicted jet structure, vortical regions and recirculation within the scour cavity.

The present study prescribed the experimentally measured equilibrium scour geometry as a fixed bed and therefore did not simulate sediment motion or bed evolution. A logical extension would be to couple the hydrodynamic model with sediment-transport and morphodynamic modelling. This would allow the development of the scour hole to be predicted directly from the computed hydraulic loading and would provide a more complete assessment of the relationship between bed shear stress and scour evolution.

Further numerical sensitivity studies are also recommended, particularly for mesh resolution

in the PKW and scour regions, downstream domain length, time-step selection and turbulence modelling. Higher-fidelity turbulence approaches such as Large Eddy Simulation could additionally be considered for detailed investigation of the strongly unsteady vortical structures, although this would require substantially greater computational resources.

Finally, future investigations should consider additional discharges, tailwater conditions, sediment sizes and PKW geometries. Such studies would help determine whether the trends observed for the present LOPKW and NLOPKW configurations remain valid over a broader range of hydraulic operating conditions and would provide a stronger basis for predicting downstream scour and designing suitable bed-protection measures.

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